

The Gittins-Index Design Principle for Cost-aware Bayesian Decision-Making under Uncertainty

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B-exam @ Cornell ORIE

Dissertation Work Collaborators



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NYU

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UCL

Raul Astudillo
MBZUAI

Motivation: Design Selection under Uncertainty

ML model training:

Training hyperparameters
(e.g., learning rate, # layers)



Accuracy

Goal: find hyperparameters that achieve high accuracy

Motivation: Design Selection under Uncertainty

ML model training:

Training hyperparameters
(e.g., learning rate, # layers)

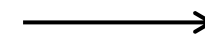
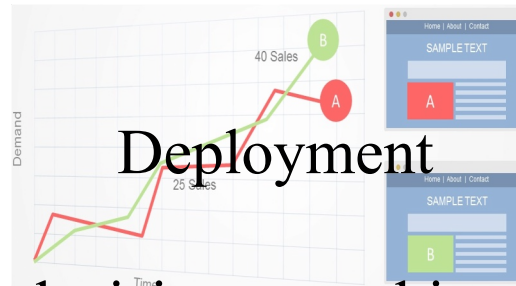


Accuracy

Goal: find hyperparameters that achieve high accuracy

Adaptive experimentation:

Decision/design variables
(e.g., UI design, pricing)



Revenue

Goal: make decisions to achieve high revenue

Motivation: Design Selection under Uncertainty



$$\text{Goal: } \max_{x \in \mathcal{X}} f(x)$$

ML model training:



Adaptive experimentation:



Motivation: Design Selection under Uncertainty

Black-box optimization:

(gradient-based methods not applicable)

Input x $\longrightarrow$



non-analytical &
no gradient info

$\longrightarrow$

Observed outcome $f(x)$

Goal: $\max_{x \in \mathcal{X}} f(x)$

ML model training:

Training hyperparameters
(e.g., learning rate, # layers)

$\longrightarrow$



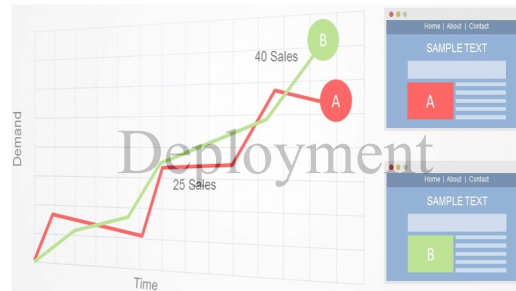
$\longrightarrow$

Accuracy

Adaptive experimentation:

Decision/design variables
(e.g., UI design, pricing)

$\longrightarrow$



$\longrightarrow$

Revenue

Motivation: Costly Evaluation for Design Selection

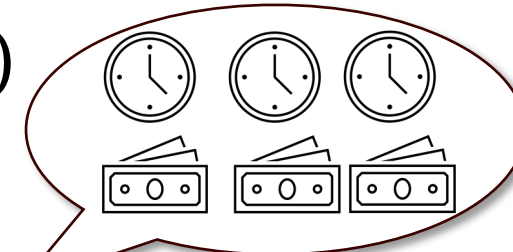
Black-box optimization:



Goal: $\max_{x \in \mathcal{X}} f(x)$

ML model training:

Training hyperparameters
(e.g., learning rate, # layers)

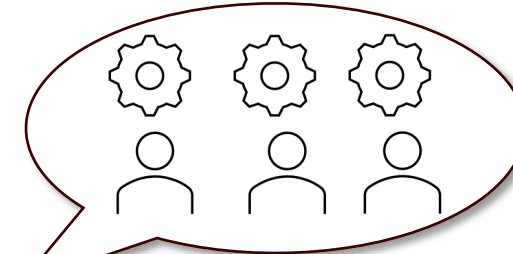
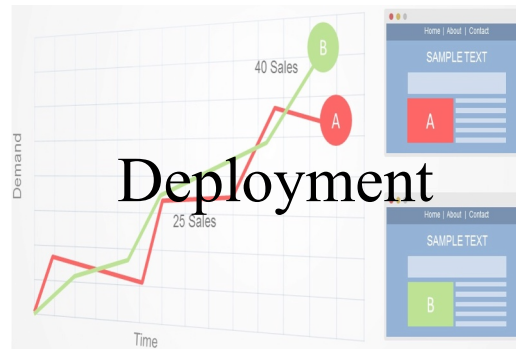


Training time
Compute credits

Accuracy

Adaptive experimentation:

Decision/design variables
(e.g., UI design, pricing)

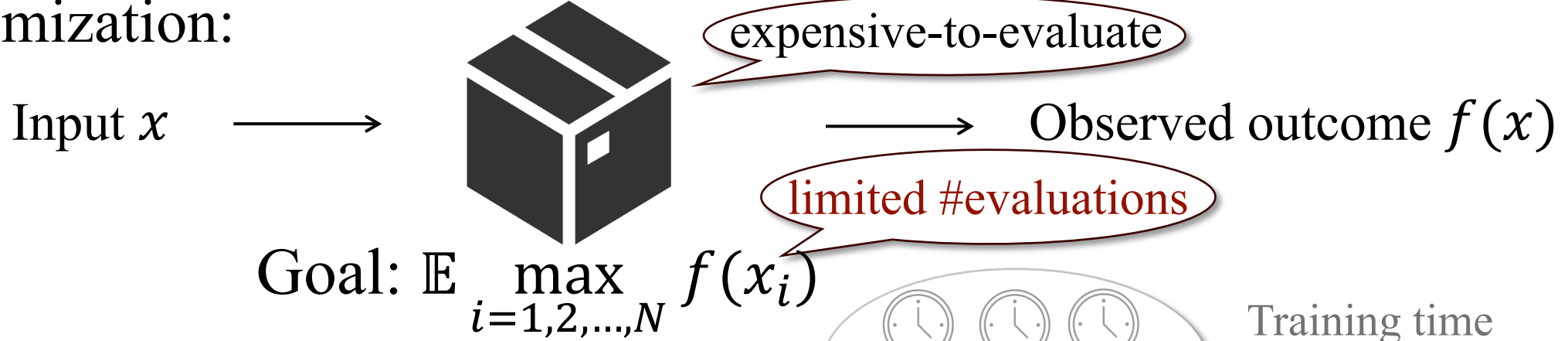


Operational cost
User experience

Revenue

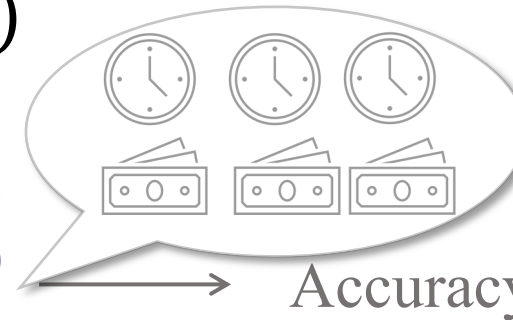
Motivation: Costly Evaluation for Design Selection

Black-box optimization:



ML model training:

Training hyperparameters
(e.g., learning rate, # layers)

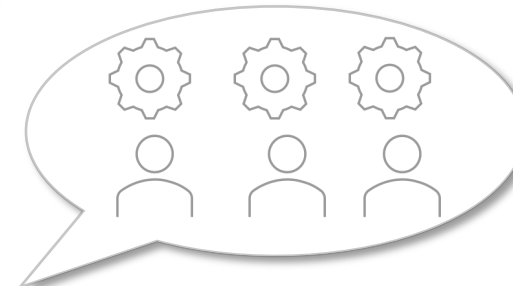
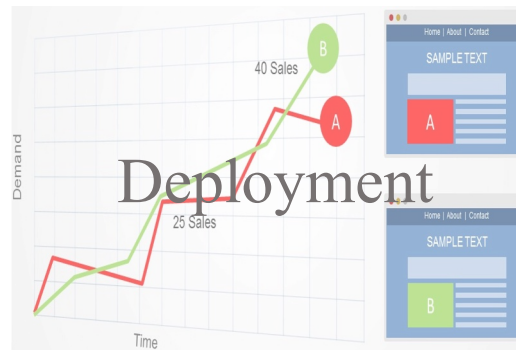


Training time
Compute credits

Accuracy

Adaptive experimentation:

Decision/design variables
(e.g., UI design, pricing)



Operational cost
User experience

Revenue

Naïve Approach: Grid Search

ML model training:

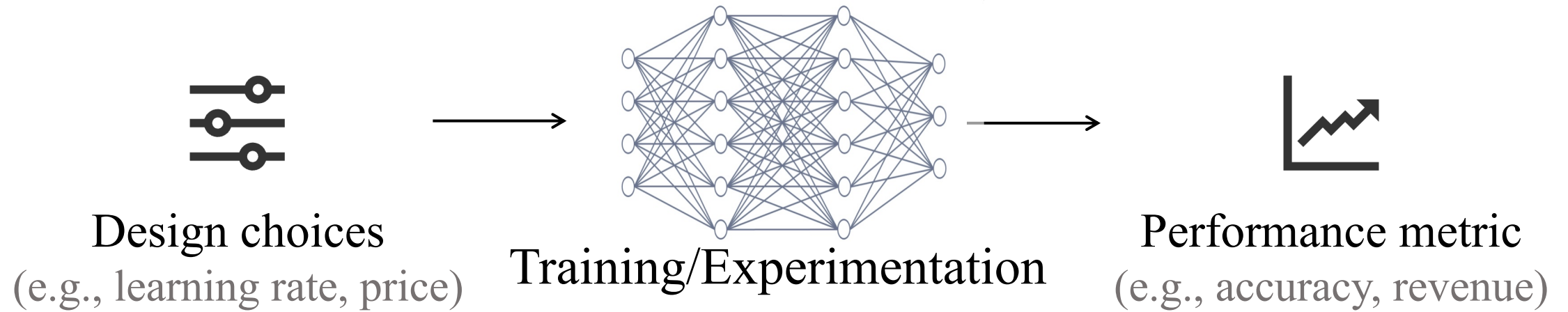


Goal: find hyperparameters that achieve high accuracy

Training hyperparameter	Range	Number of Options
Batch size	[16, 512]	10
Learning rate	[1e-4, 1e-1]	10
Momentum	[0.1, 0.99]	10
Weight decay	[1e-5, 1e-1]	10
Number of layers	{1, 2, 3, 4}	4
Max units per layer	[64, 1024]	10
Dropout	[0.0, 1.0]	10

40,000,000
combinations!

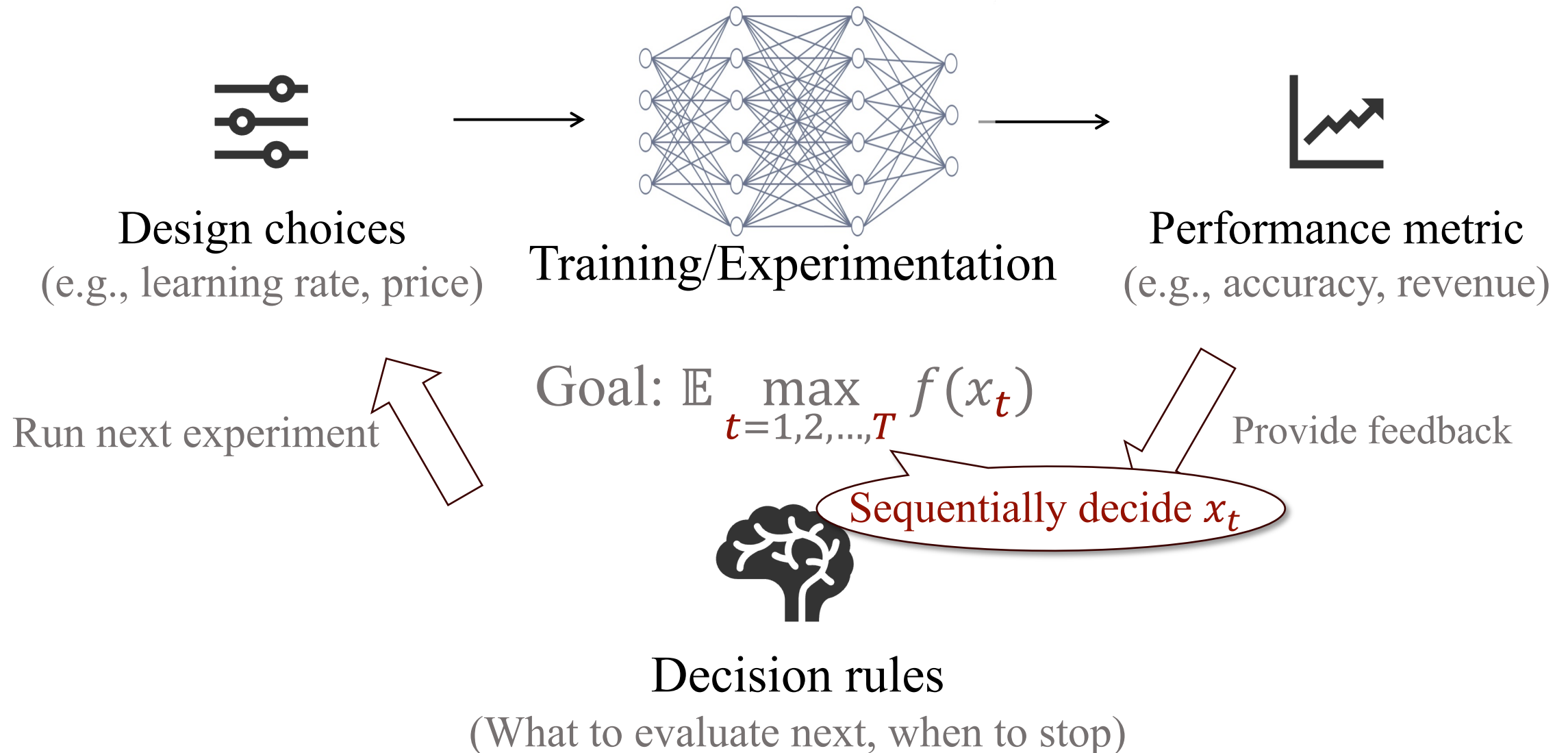
Sample-Efficient Decision-Making



$$\text{Goal: } \mathbb{E} \max_{i=1,2,\dots,N} f(x_i)$$

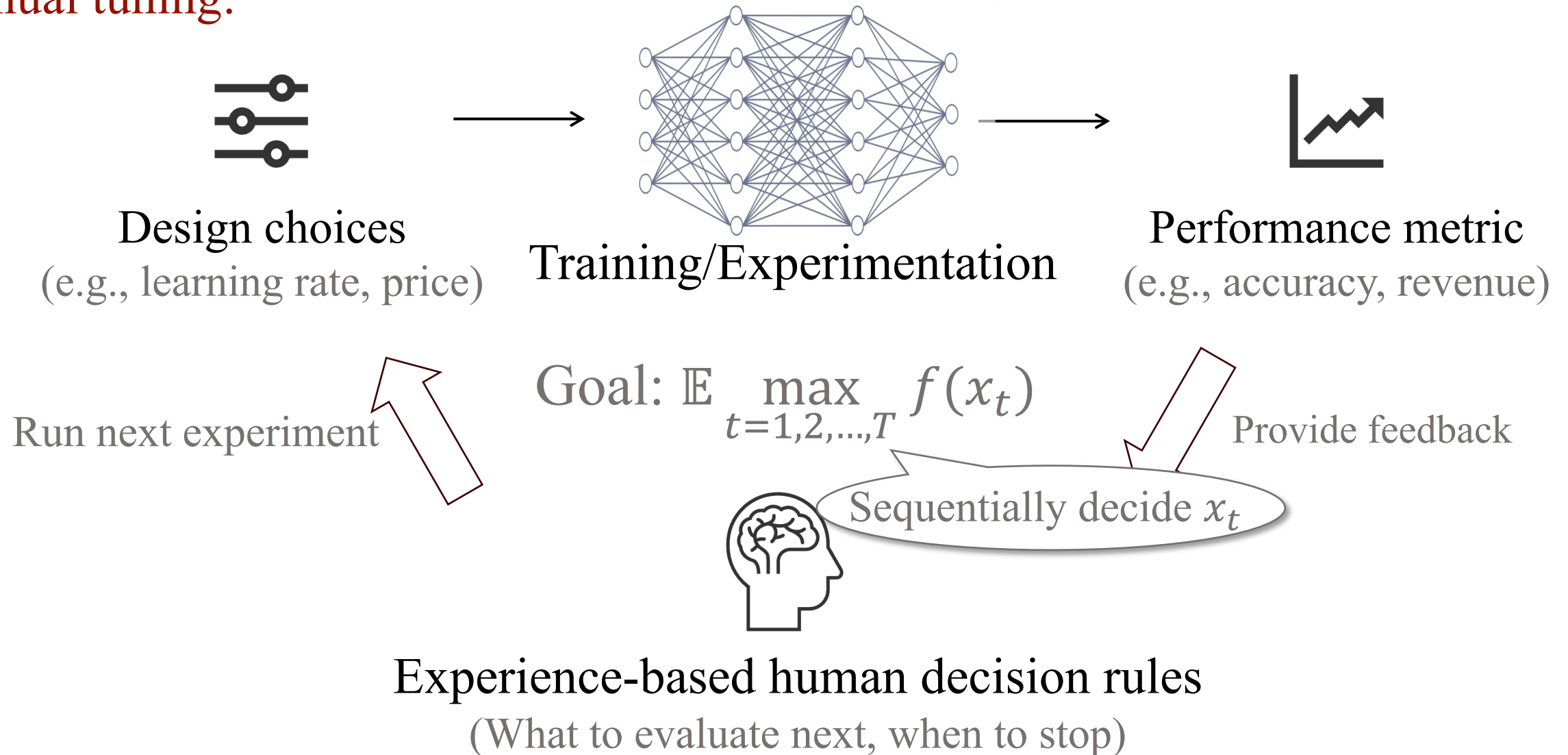
As few as possible

Sample-Efficient Adaptive Decision-Making



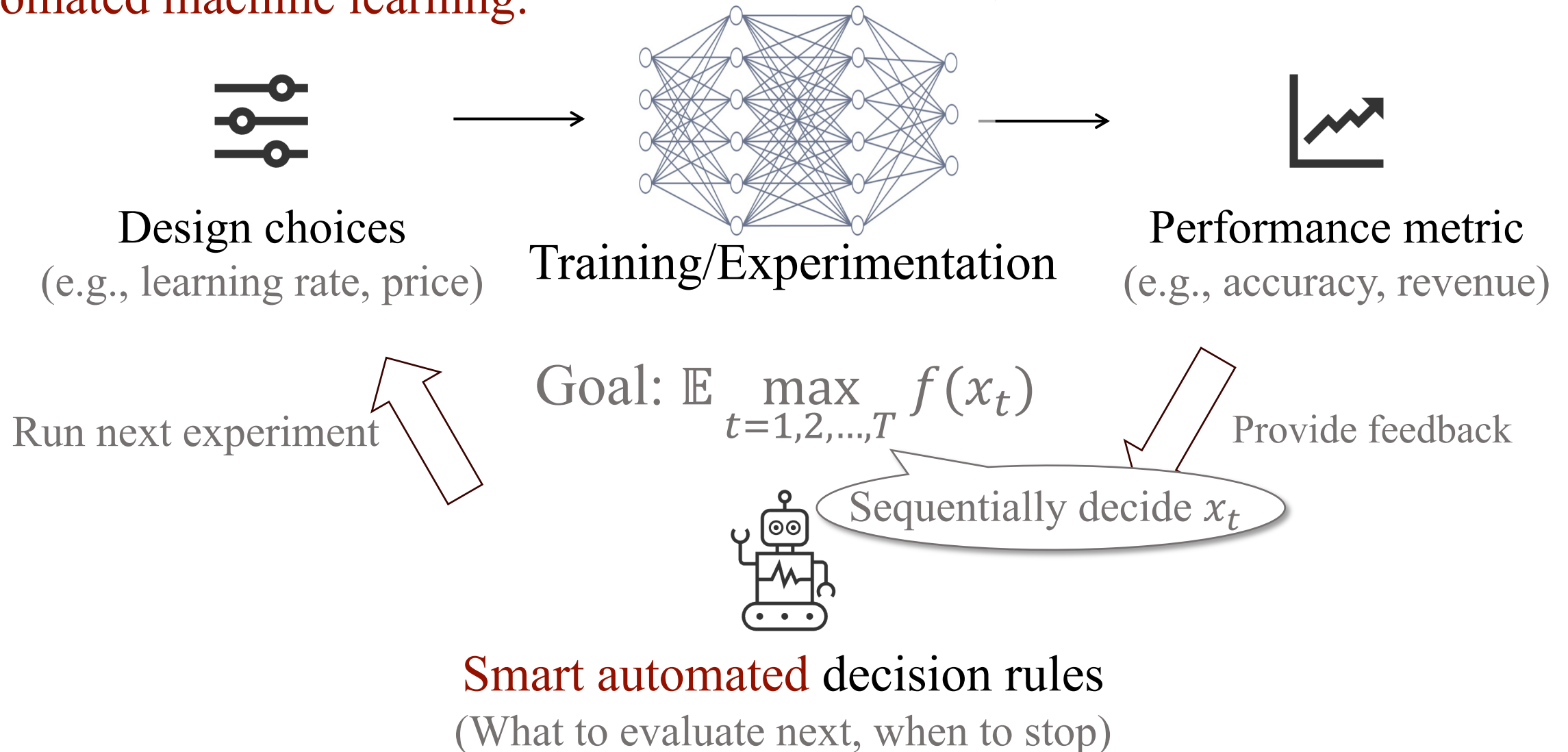
Sample-Efficient Adaptive Decision-Making

Manual tuning:



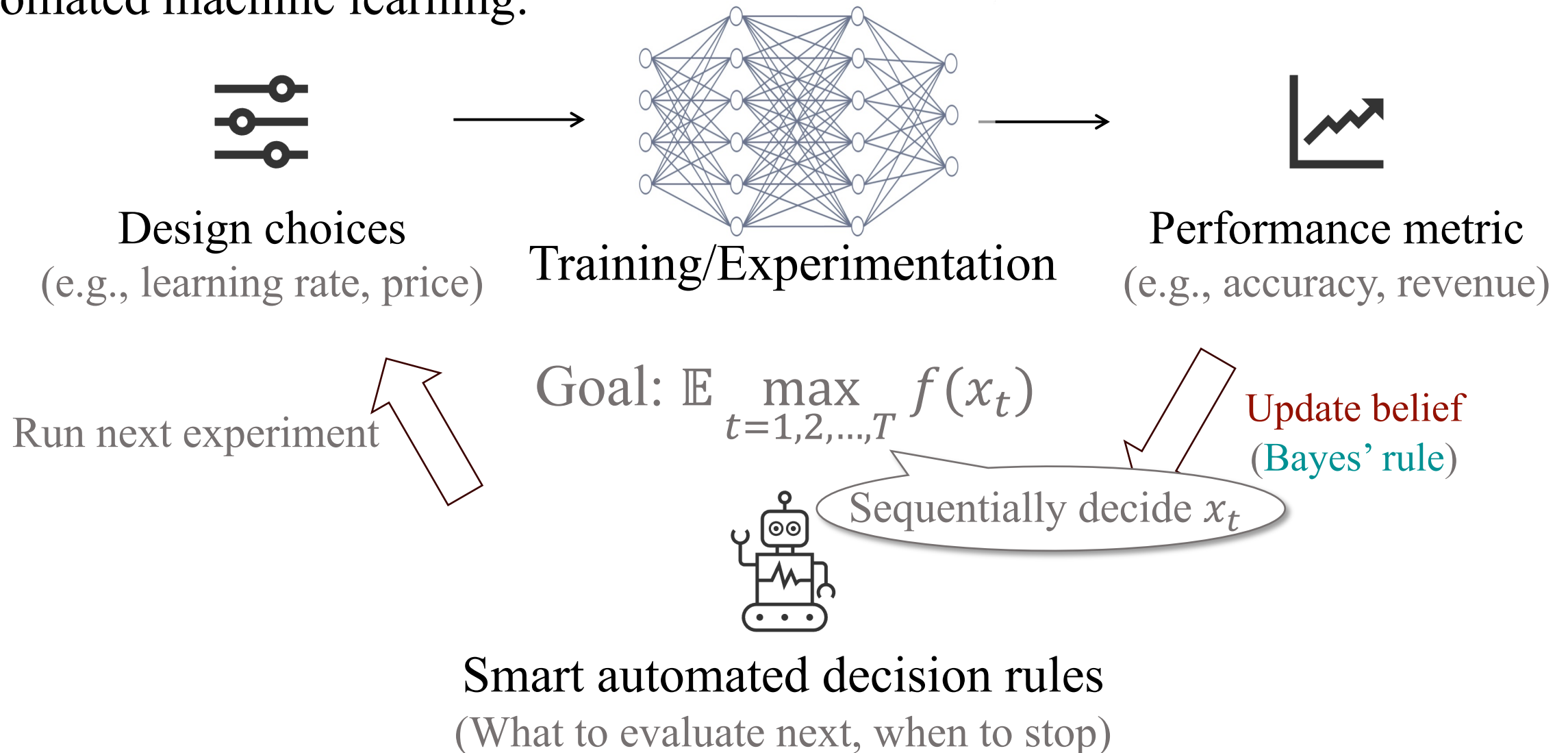
Sample-Efficient Adaptive Decision-Making

Automated machine learning:

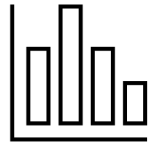


Sample-Efficient Bayesian Decision-Making

Automated machine learning:



Under-explored Factors for Better Decisions



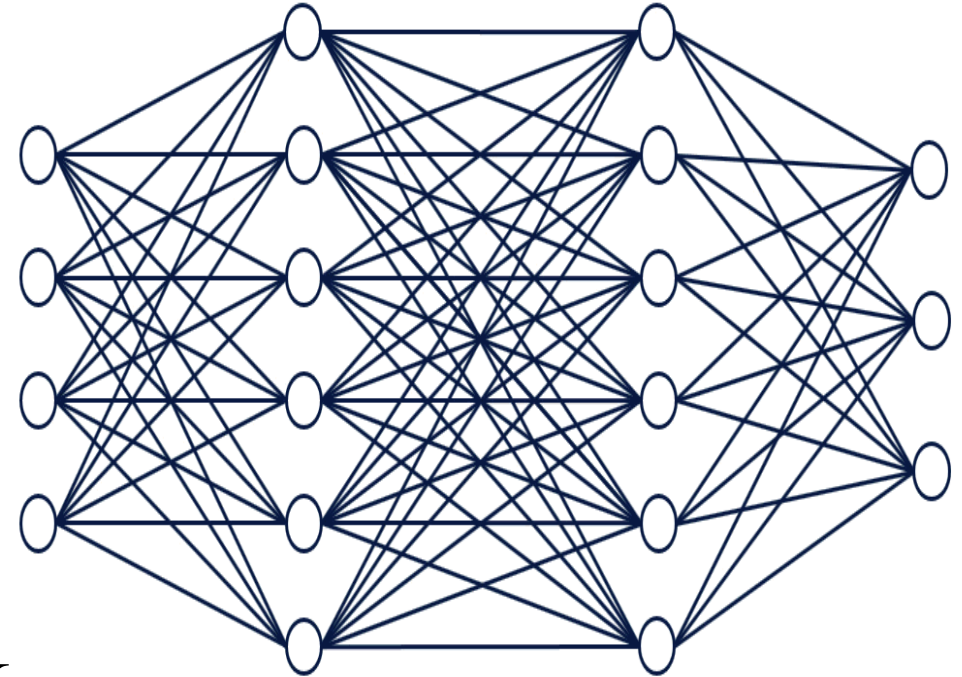
Varying evaluation costs



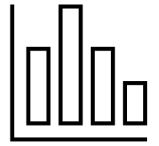
Smart stopping time



Observable multi-stage feedback



Under-explored Factors for Better Decisions



Varying evaluation costs



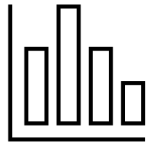
Smart stopping time



Observable multi-stage feedback

New design principle:
Gittins index

Why Gittins index?



Varying evaluation costs



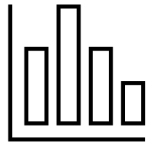
Smart stopping time



Observable multi-stage feedback

Featured in related sequential
decision problems

Why Gittins index?



Varying evaluation costs

Featured in **Pandora's box**



Smart stopping time

Featured in **Pandora's box**



Observable multi-stage feedback

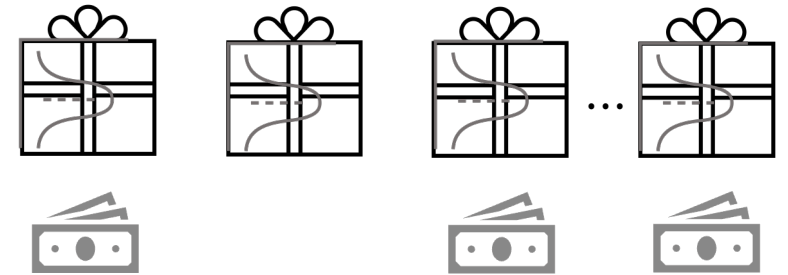


"Cost-aware Bayesian Optimization via the Pandora's Box Gittins Index." NeurIPS'24.

Part I

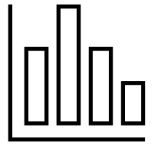
New design principle:
Gittins index

Featured in related sequential
decision problems



"Cost-aware Stopping for Bayesian Optimization." ICML'26.

Why Gittins index?



Varying evaluation costs

Featured in Pandora's box



Smart stopping time

Featured in Pandora's box



Observable multi-stage feedback

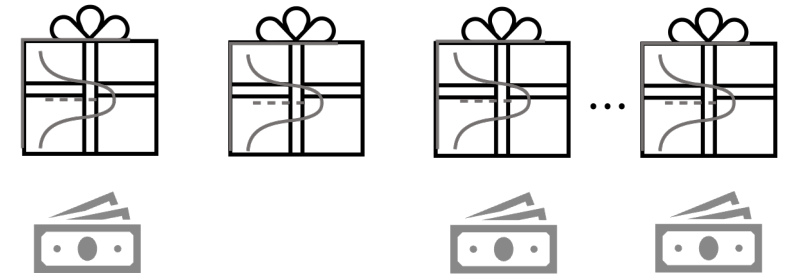


"Cost-aware Bayesian Optimization via the Pandora's Box Gittins Index." NeurIPS'24.

Part I

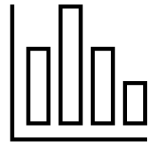
New design principle:
Gittins index

Optimal in related sequential
decision problems



"Cost-aware Stopping for Bayesian Optimization." ICML'26.

Why Gittins index?



Varying evaluation costs

Features in Pandora's box

Part I



Smart stopping time

Features in Pandora's box



Observable multi-stage feedback

Featured in **Markov chain selection**

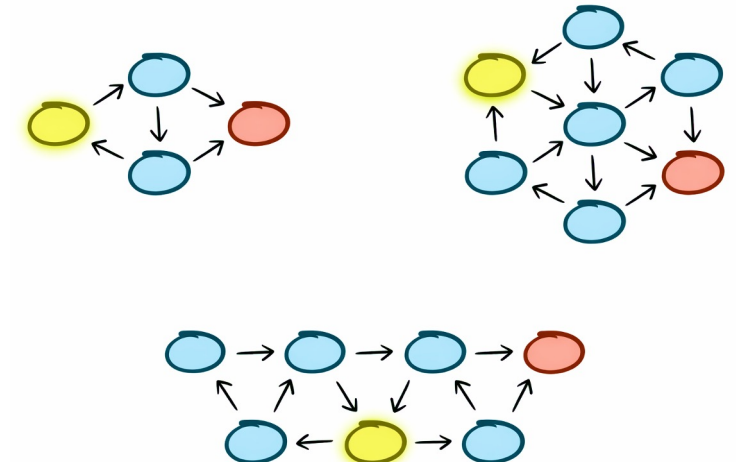
Part II



"Efficient Cost-Aware LLM Evaluation via Bayesian Bandit Gittins Indices." ICML'26 Workshop DEMO.

New design principle:
Gittins index

Optimal in related sequential
decision problems



Talk Outline

Part I:

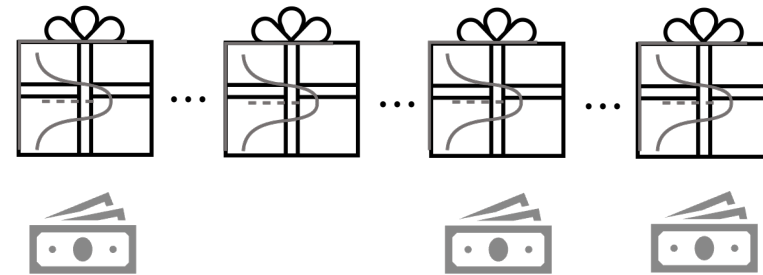
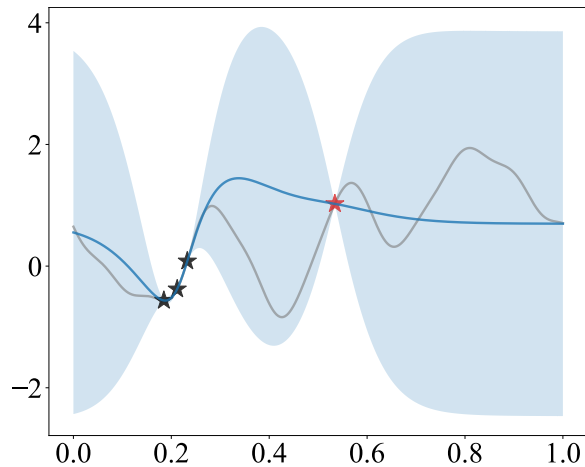
Cost-aware Bayesian Optimization with Adaptive Stopping via Pandora's Box Gittins Indices

Part II:

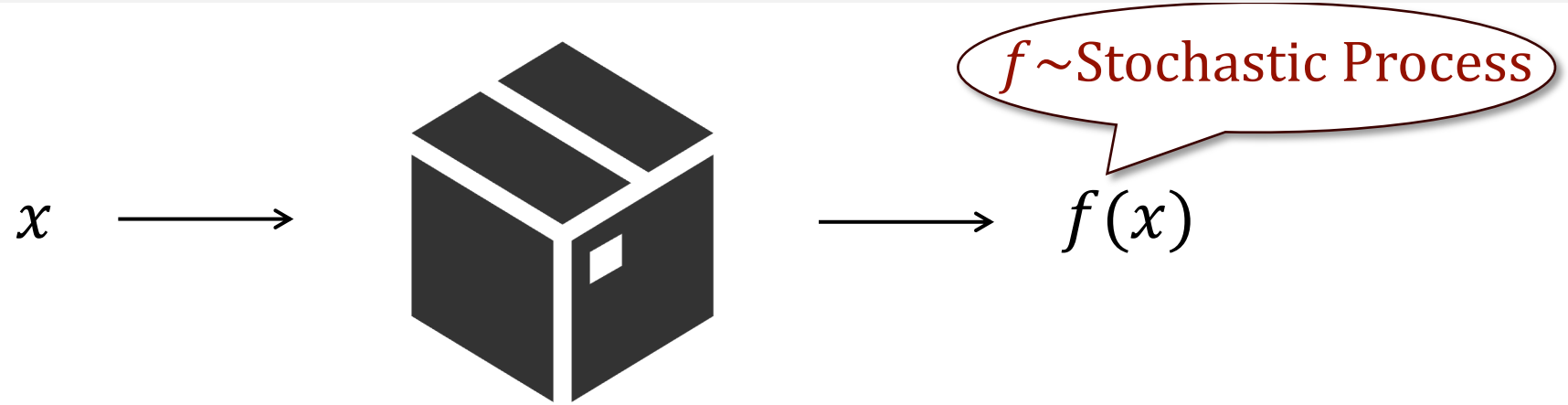
Towards Bayesian Optimization with Multi-Stage Feedback via Markov Chain Gittins Indices

Part I:

Cost-aware Bayesian Optimization with Adaptive Stopping via Pandora's Box Gittins Indices



Bayesian Optimization

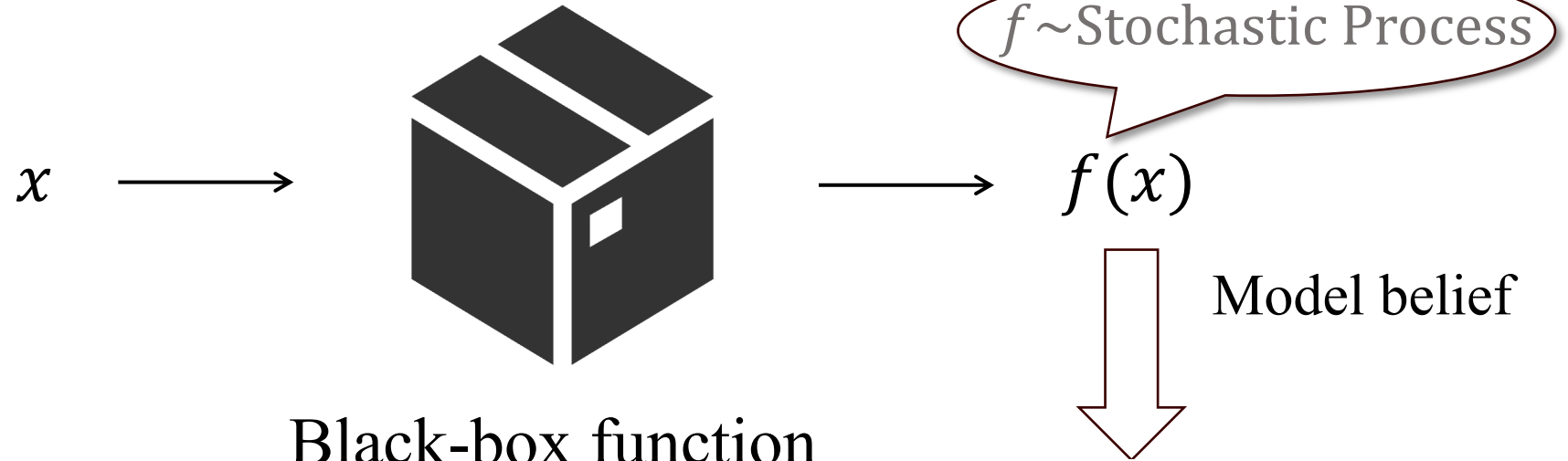


Black-box function

$$\text{Goal: } \mathbb{E} \max_{t=1,2,\dots,T} f(x_t)$$

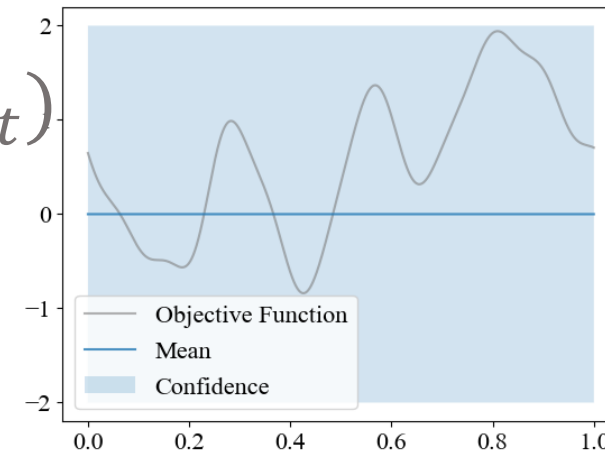
Bayesian Optimization

Time 0



Black-box function

Goal: $\mathbb{E} \max_{t=1,2,\dots,T} f(x_t)$



Probabilistic model
(e.g., Gaussian process)

Bayesian Optimization

Time t

$x_1, \dots, x_t$



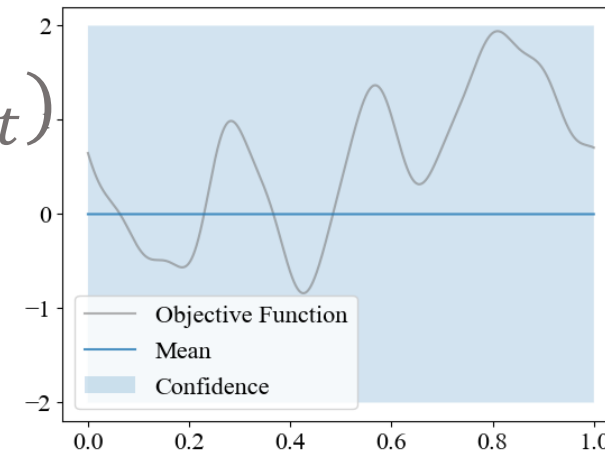
$f \sim \text{Stochastic Process}$

$f(x_1), \dots, f(x_t)$

Model belief

Black-box function

Goal: $\mathbb{E} \max_{t=1,2,\dots,T} f(x_t)$



Probabilistic model
(e.g., Gaussian process)

Bayesian Optimization

Time t

$x_1, \dots, x_t$



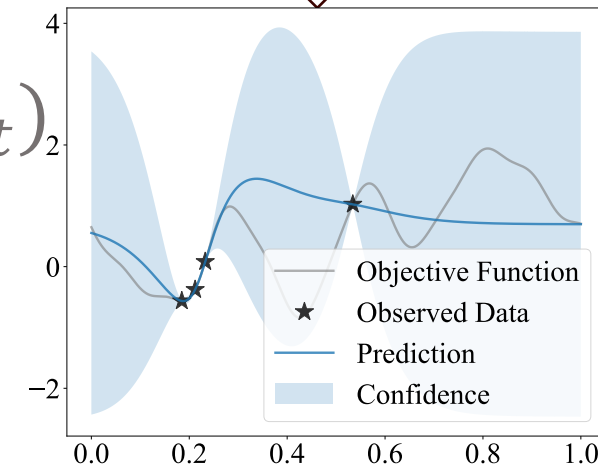
$f \sim \text{Stochastic Process}$

$f(x_1), \dots, f(x_t)$

Update belief
(Bayes' rule)

Black-box function

Goal: $\mathbb{E} \max_{t=1,2,\dots,T} f(x_t)$

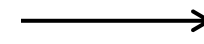
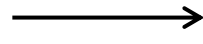


Probabilistic model
(e.g., Gaussian process)

Bayesian Optimization

Time t

$x_1, \dots, x_t$



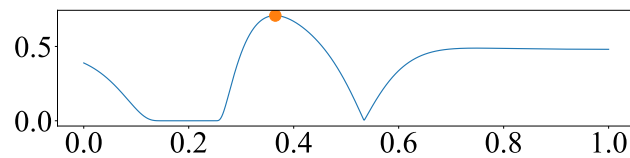
$f(x_1), \dots, f(x_t)$

Update belief
(Bayes' rule)



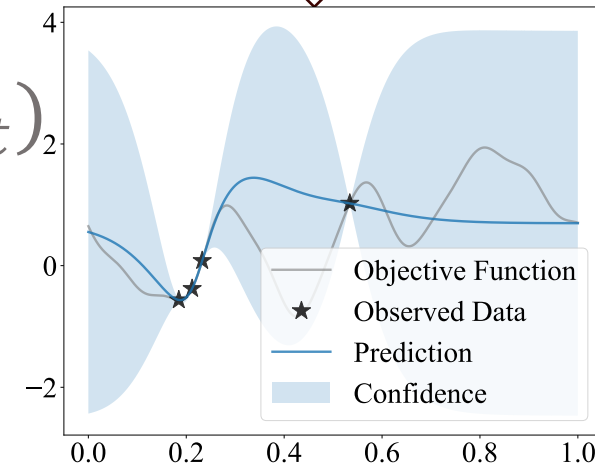
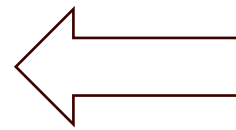
Black-box function

Goal: $\mathbb{E} \max_{t=1,2,\dots,T} f(x_t)$



Decision rule
(What to evaluate next)

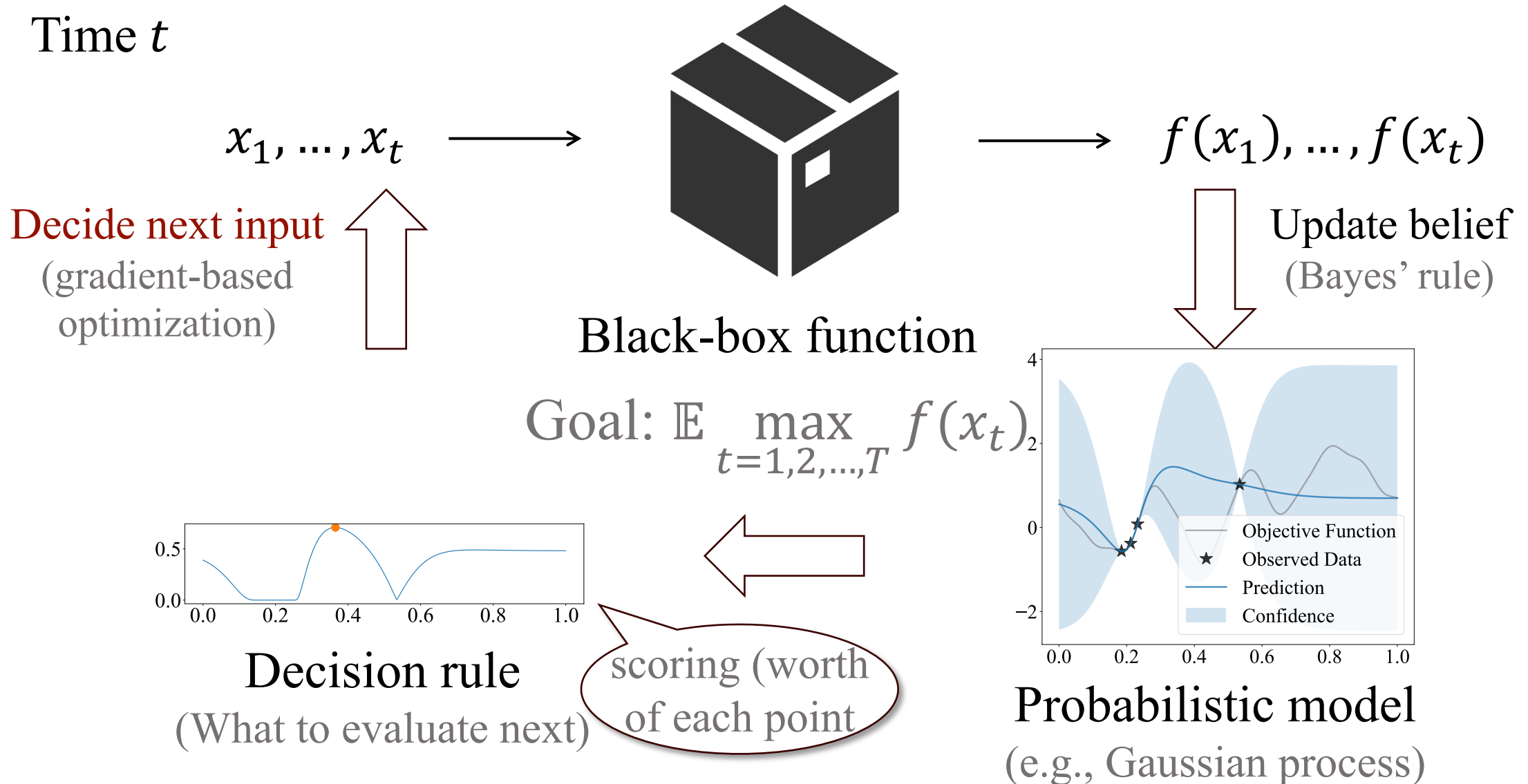
scoring (worth
of each point)



Probabilistic model
(e.g., Gaussian process)

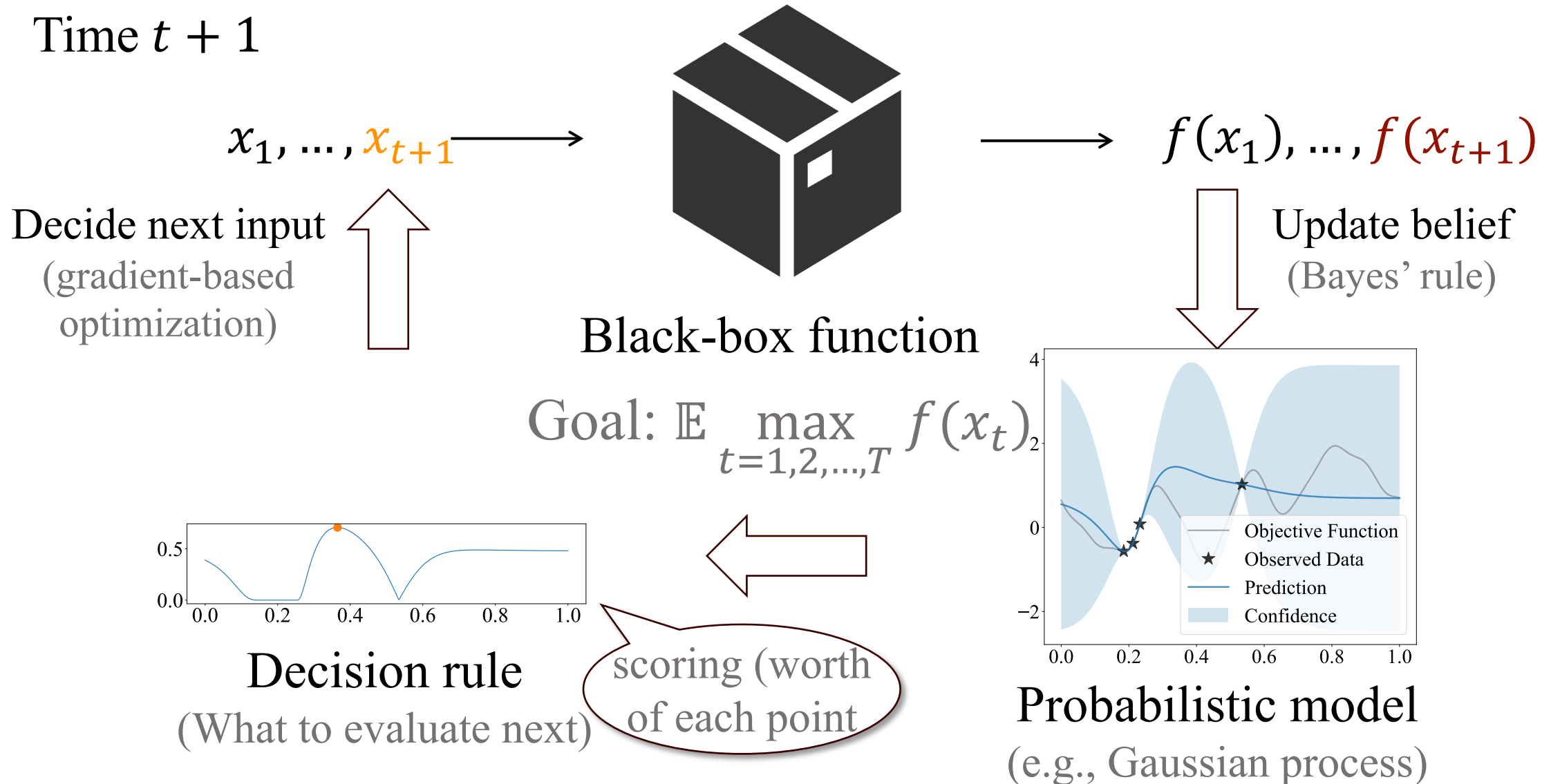
Bayesian Optimization

Time t



Bayesian Optimization

Time $t + 1$



Bayesian Optimization

Time $t + 1$

$x_1, \dots, x_{t+1}$



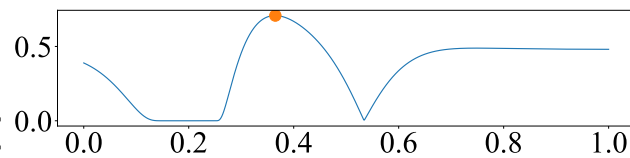
$f(x_1), \dots, f(x_{t+1})$

Decide next input
(gradient-based
optimization)



Black-box function

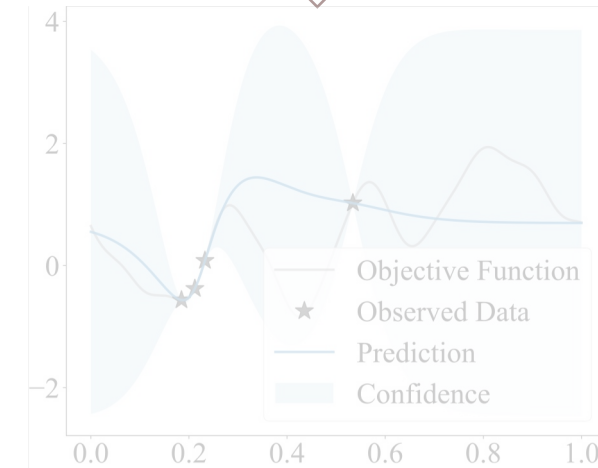
Update belief
(Bayes' rule)



My focus

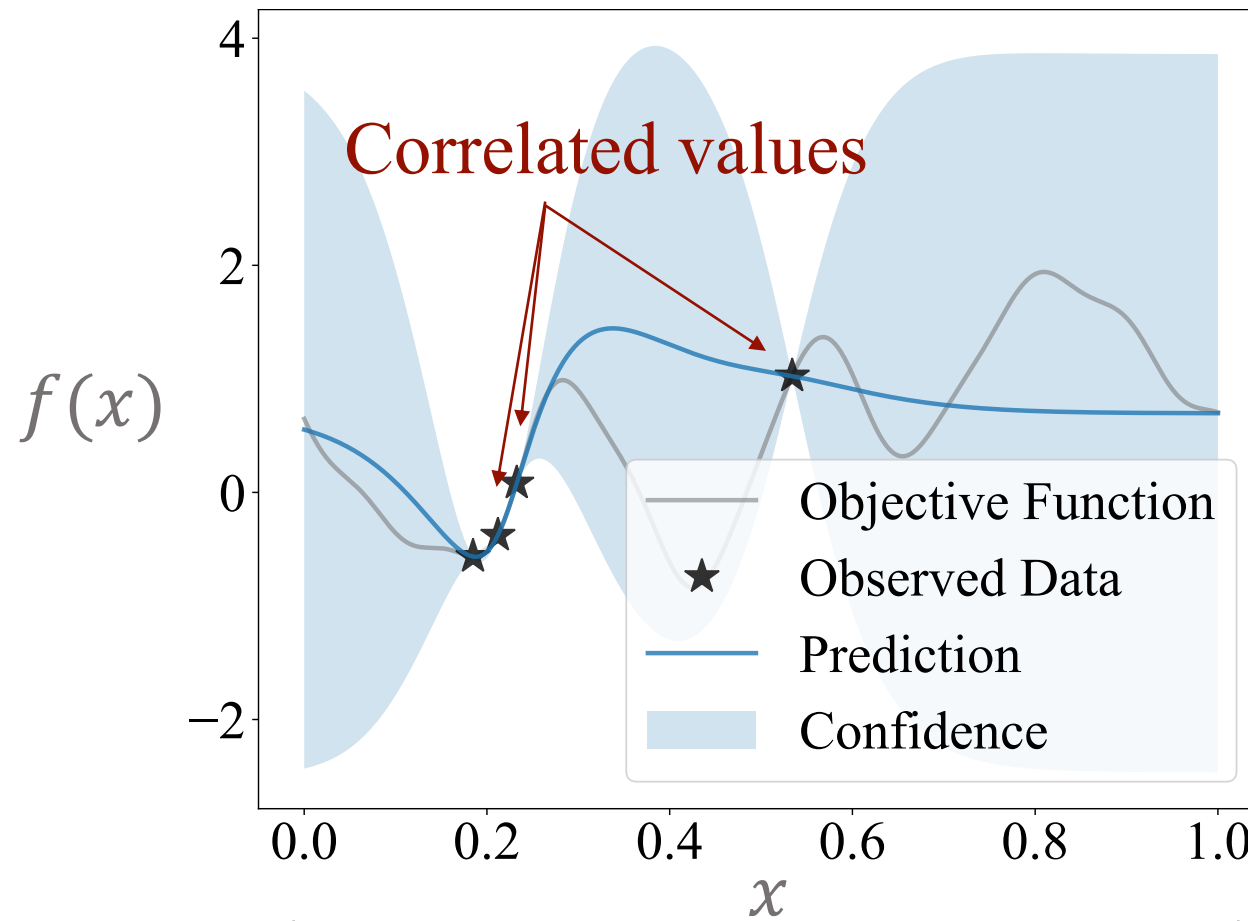
Decision rule
(What to evaluate next)

scoring (worth
of each point)



Probabilistic model
(e.g., Gaussian process)

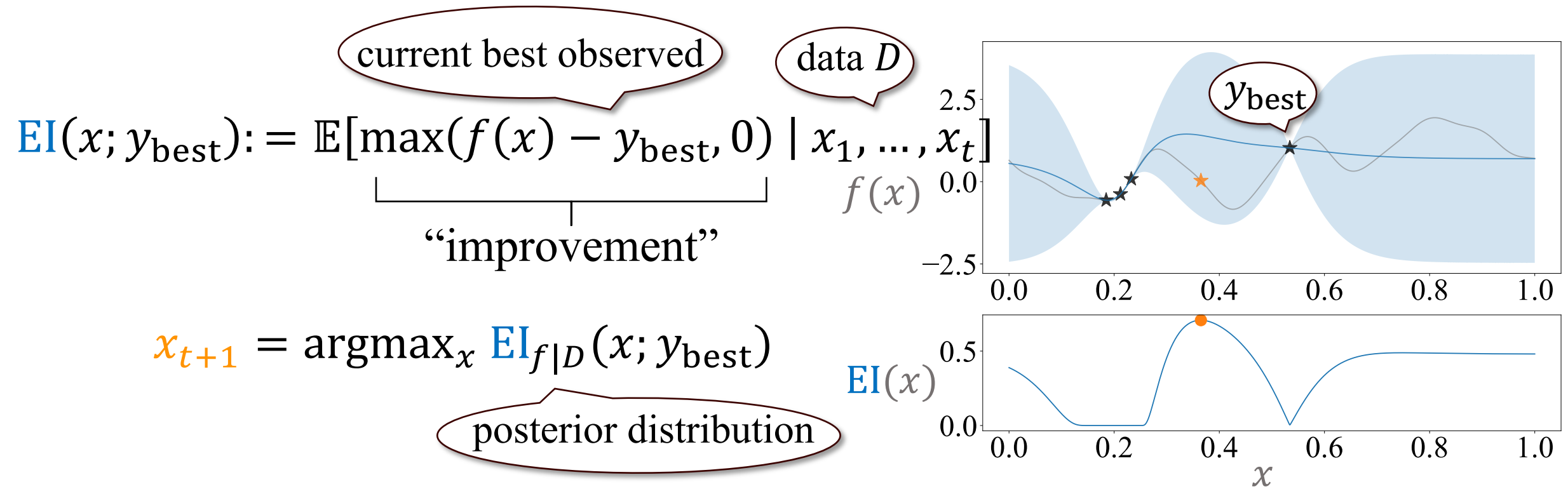
Challenges in Decision Rule Design



Continuous search space

Correlation & continuity $\Rightarrow$ Intractable MDP $\Rightarrow$ Optimal policy unknown

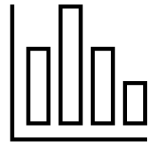
Popular Decision Rule: Expected Improvement



One-step approximation to MDP

Expected improvement

How does EI incorporate costs and stopping?



Varying evaluation costs

$$\text{EIPC}_c(x; y_{\text{best}}) = \text{EI}(x; y_{\text{best}}) / c(x)$$

[Snoek et al.'12]

Arbitrarily bad

[Astudillo et al.'21]

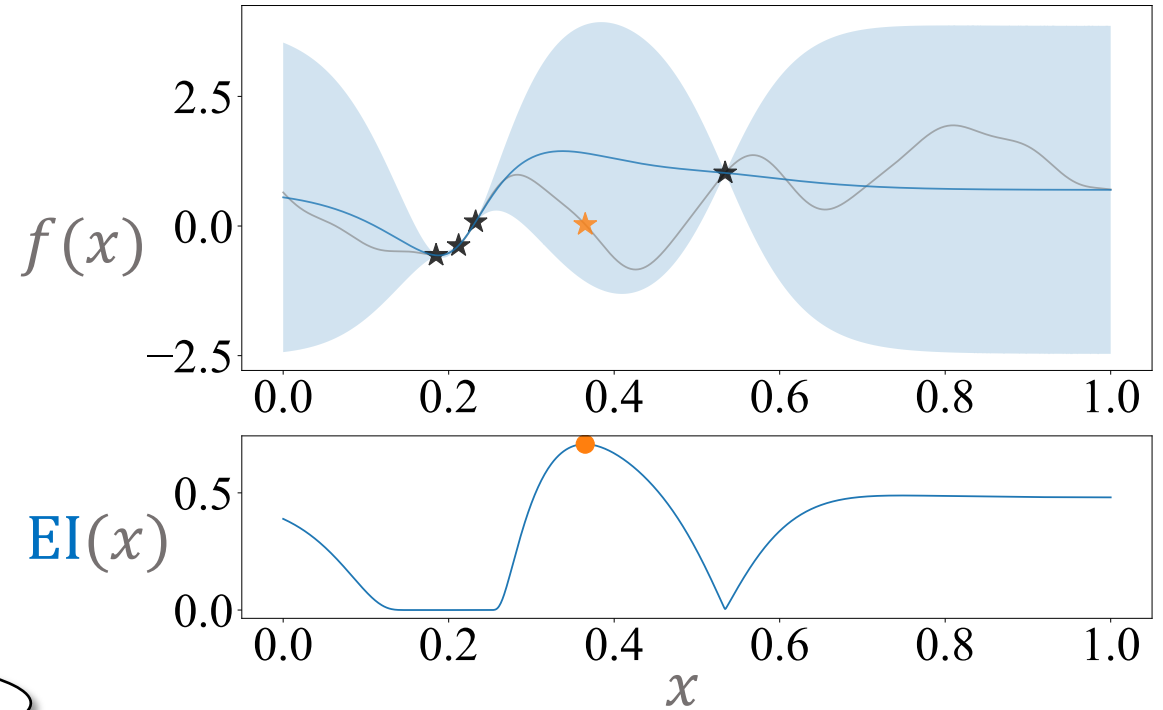


Smart stopping time

$$\tau: \text{EI}(x_\tau; y_{\text{best}}) \leq \theta$$

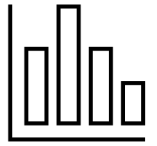
[Locatelli'97,
Nguyen et al.'17,
Ishibashi et al.'23]

Which threshold?



Expected improvement

Expected Improvement vs Gittins Index



Varying evaluation costs

$$\text{EIPC}_c(x; y_{\text{best}}) = \text{EI}(x; y_{\text{best}})/c(x)$$

Arbitrarily bad

$$\text{GI}_c(x)$$

Costs built-in



Smart stopping time

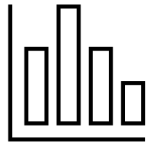
$$\tau: \text{EI}(x_\tau; y_{\text{best}}) \leq \theta$$

Which threshold?

$$\tau: \text{GI}_c(x_\tau) \leq y_{\text{best}}$$

Built-in threshold

Expected Improvement vs Gittins Index



Varying evaluation costs

$$\text{EIPC}_c(x; y_{\text{best}}) = \text{EI}(x; y_{\text{best}})/c(x)$$

Arbitrarily bad

$$\text{GI}_c(x)$$

Costs built-in



Smart stopping time

$$\tau: \text{EI}(x_\tau; y_{\text{best}}) \leq \theta$$

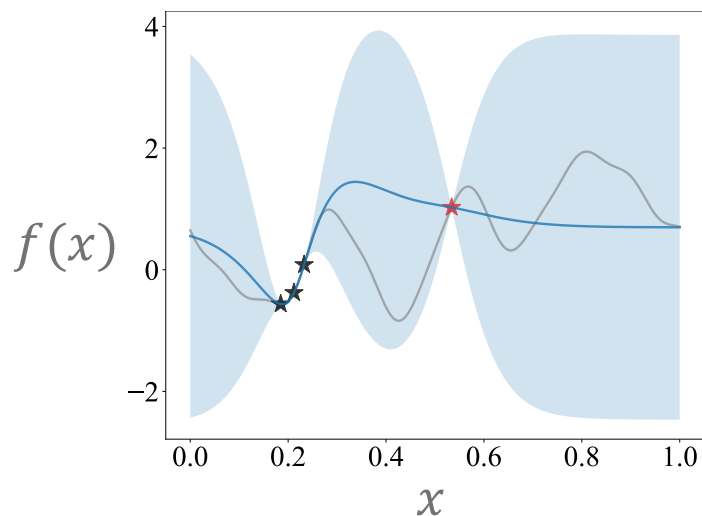
Which threshold?

$$\tau: \text{GI}_c(x_\tau) \leq y_{\text{best}}$$

Built-in threshold

Where do Gittins indices come from?

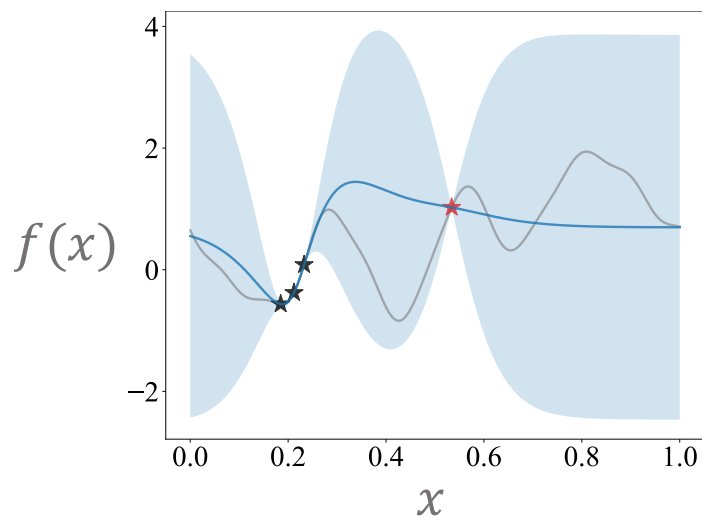
Bayesian Optimization



Continuous search space

Correlated function values

Bayesian Optimization



Continuous search space



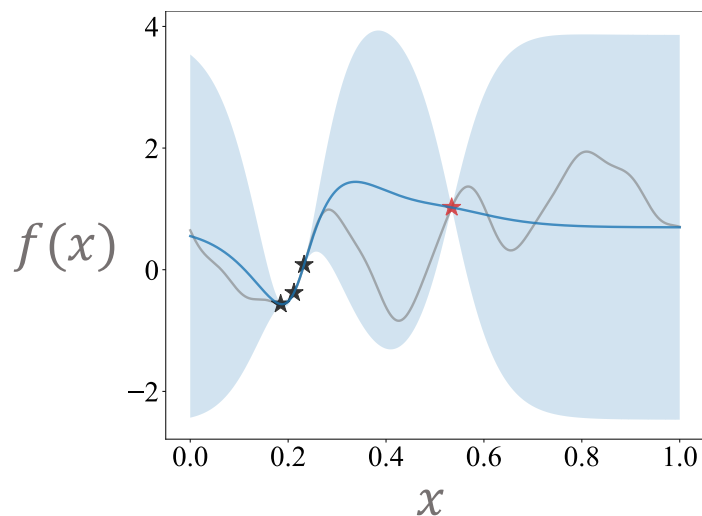
Discrete

Correlated function values



Independent

Bayesian Optimization

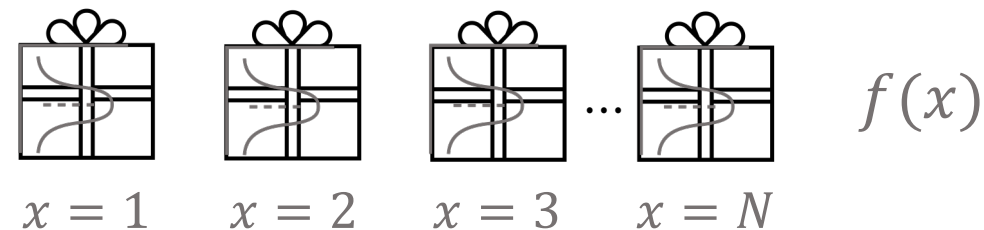


Continuous search space

Correlated function values

Pandora's Box

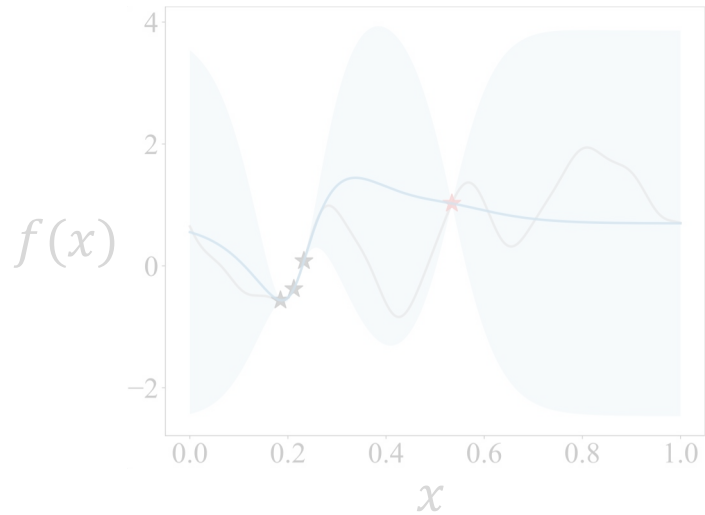
[Weitzman'79]



Discrete

Independent

Bayesian Optimization

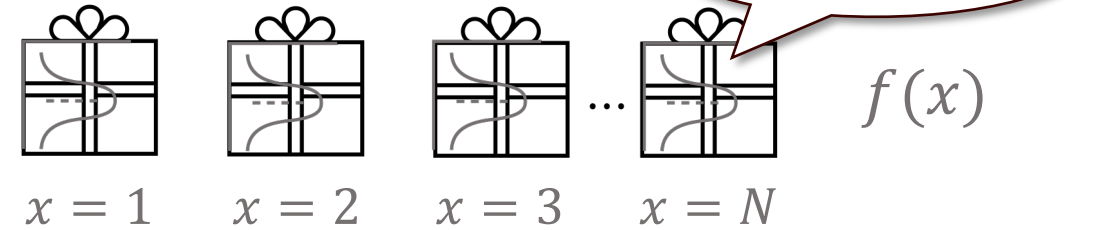


Continuous search space

Correlated function values

Pandora's Box

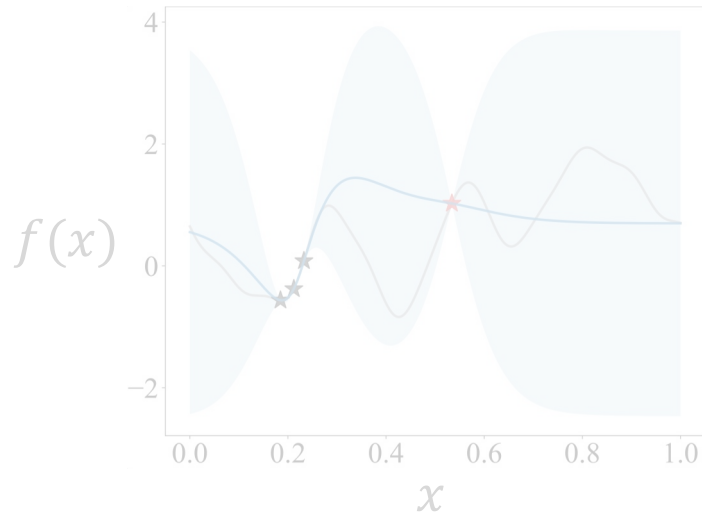
[Weitzman'79]



Discrete

Independent

Bayesian Optimization



Continuous search space

Correlated function values

Pandora's Box

[Weitzman'79]

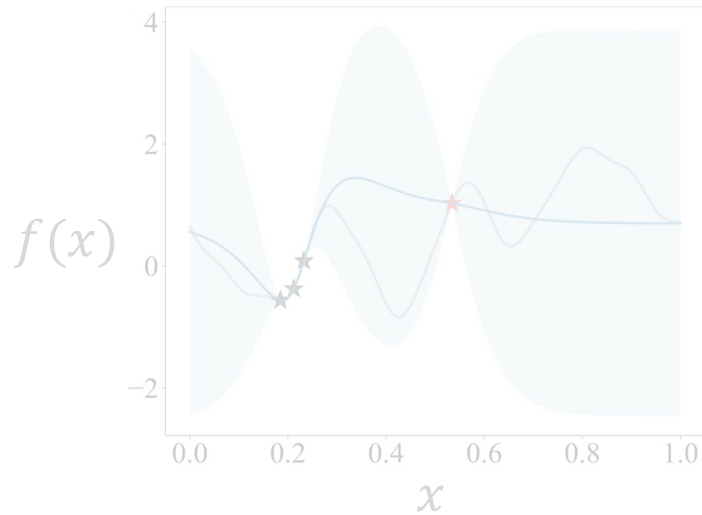


Discrete

Independent

$$\mathbb{E} \max_{t=1,2,\dots,T} f(x_t) - \mathbb{E} \sum_{t=1}^T c(x_t)$$

Bayesian Optimization

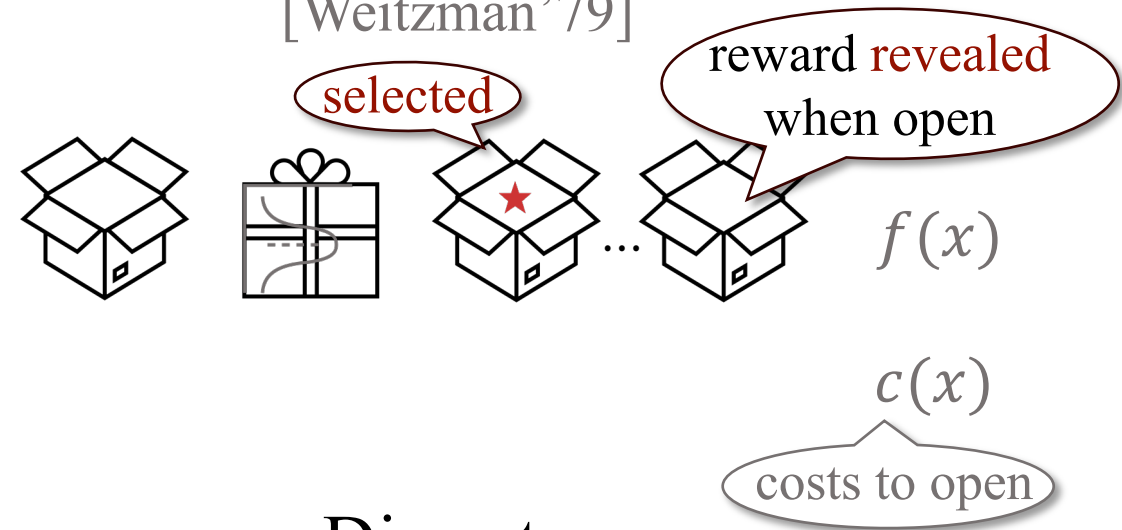


Continuous search space

Correlated function values

Pandora's Box

[Weitzman'79]

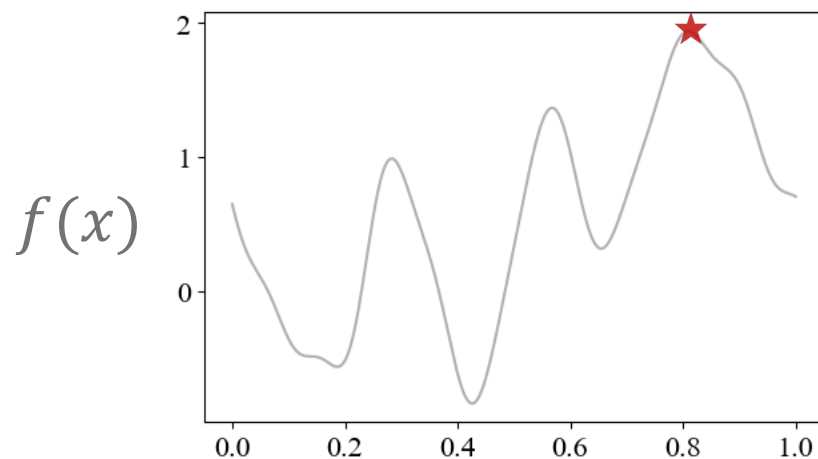


Discrete

Independent

$$\mathbb{E} \max_{t=1,2,\dots,T} f(x_t) - \mathbb{E} \sum_{t=1}^T c(x_t)$$

Bayesian Optimization



Continuous

Correlated

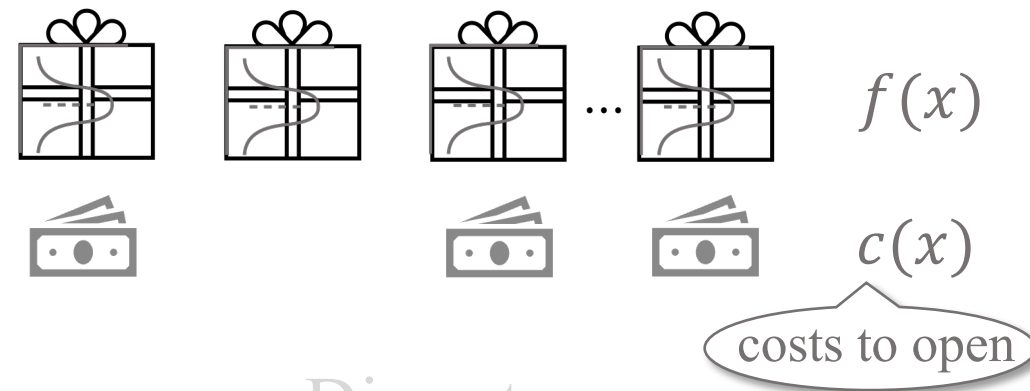
Cost-unaware

Fixed-iteration

$$\mathbb{E} \max_{t=1,2,\dots,T} f(x_t)$$

Pandora's Box

[Weitzman'79]



Discrete

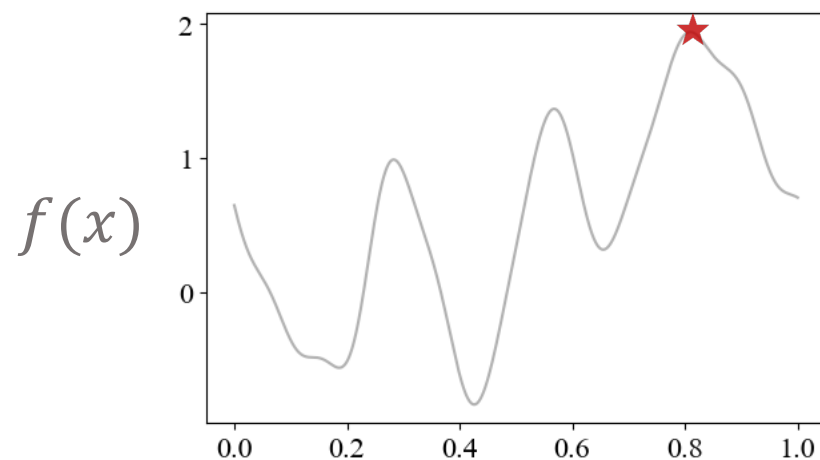
Independent

Cost-aware

Flexible-stopping

$$\mathbb{E} \max_{t=1,2,\dots,T} f(x_t) - \mathbb{E} \sum_{t=1}^T c(x_t)$$

Cost-aware Bayesian Optimization



Continuous

Correlated

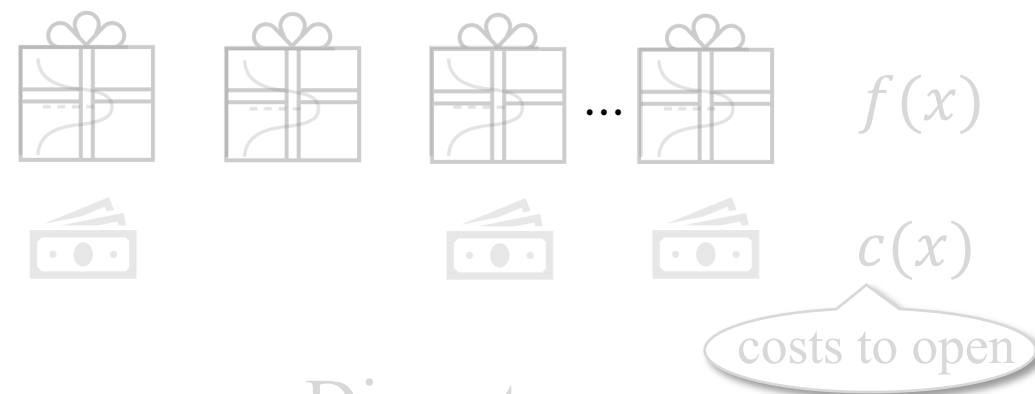
Cost-aware

Flexible-stopping

$$\mathbb{E} \max_{t=1,2,\dots,T} f(x_t) - \mathbb{E} \sum_{t=1}^T c(x_t)$$

Pandora's Box

[Weitzman'79]



Discrete

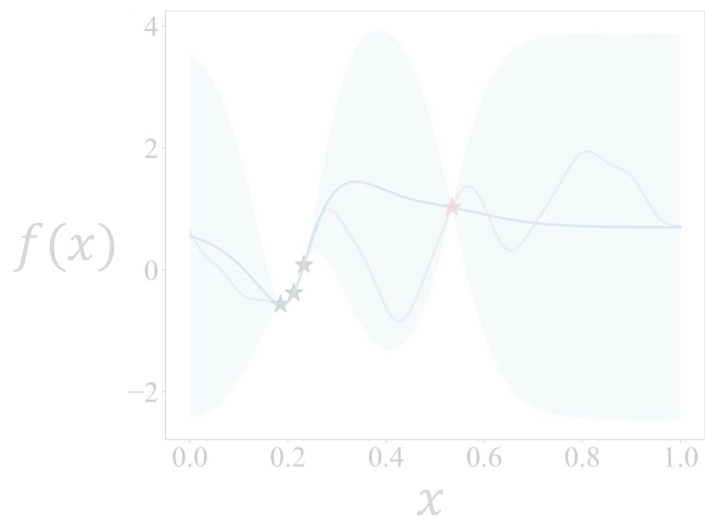
Independent

Cost-aware

Flexible-stopping

$$\mathbb{E} \max_{t=1,2,\dots,T} f(x_t) - \mathbb{E} \sum_{t=1}^T c(x_t)$$

Cost-aware Bayesian Optimization

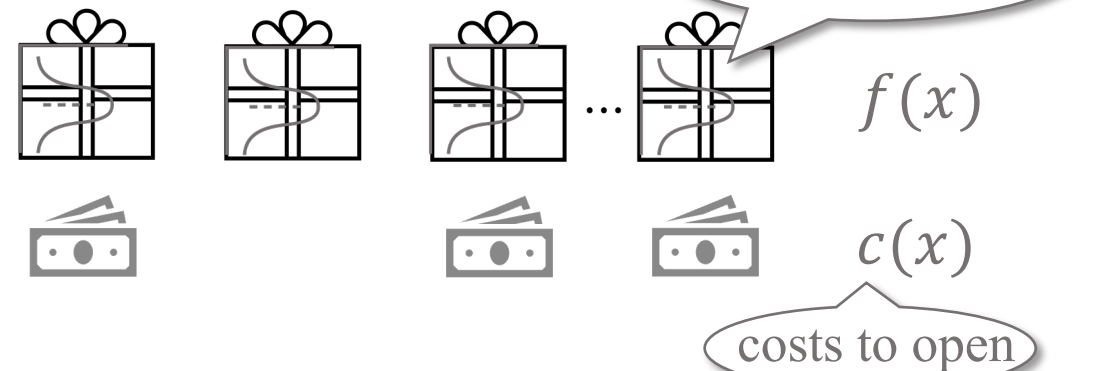


Continuous

Correlated

Pandora's Box

[Weitzman'79]

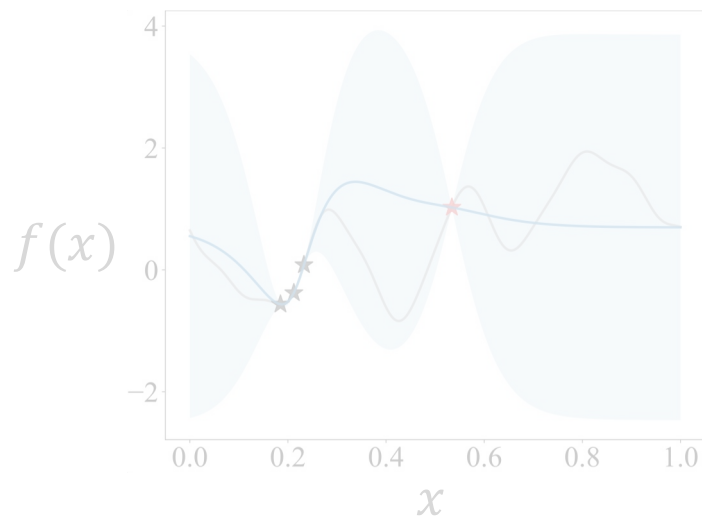


Discrete

Independent

Optimal policy: Gittins index

Cost-aware Bayesian Optimization

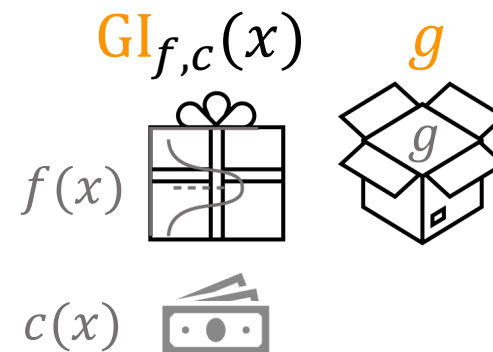


Continuous

Correlated

Pandora's Box

[Weitzman'79]



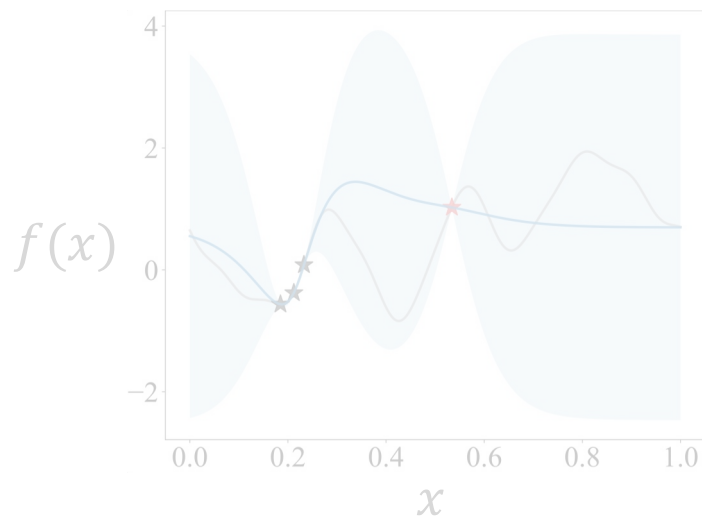
Step 1: Assign each box a Gittins index

Discrete

Independent

Optimal policy: Gittins index

Cost-aware Bayesian Optimization

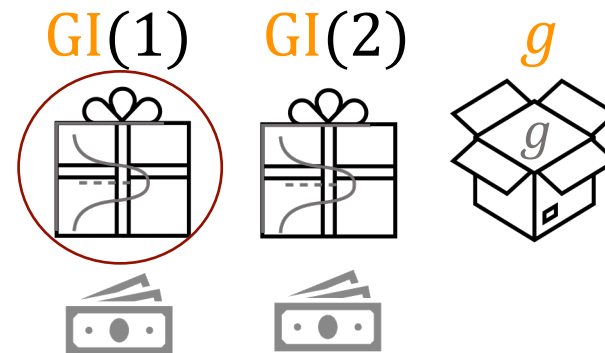


Continuous

Correlated

Pandora's Box

[Weitzman'79]



Step 2: Act on the box with the **highest** index

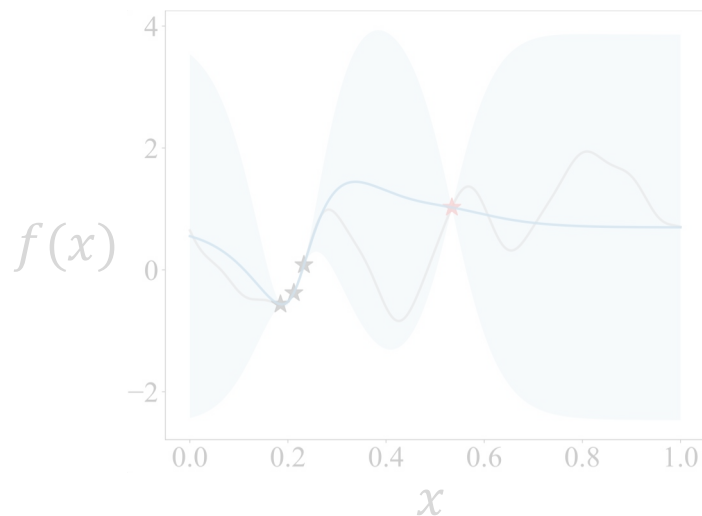
- *Closed*: **open** it
- *Opened*: select & stop

Discrete

Independent

Optimal policy: **Gittins index**

Cost-aware Bayesian Optimization

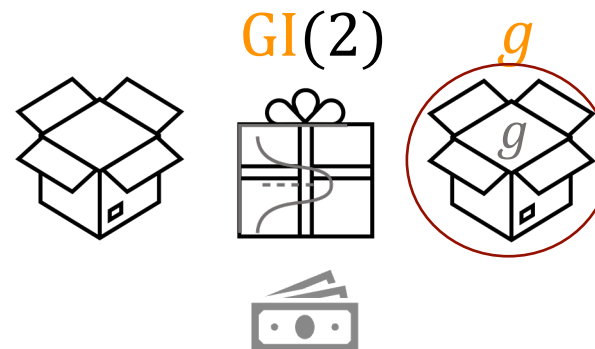


Continuous

Correlated

Pandora's Box

[Weitzman'79]



Step 2': Act on the box with the **highest** index

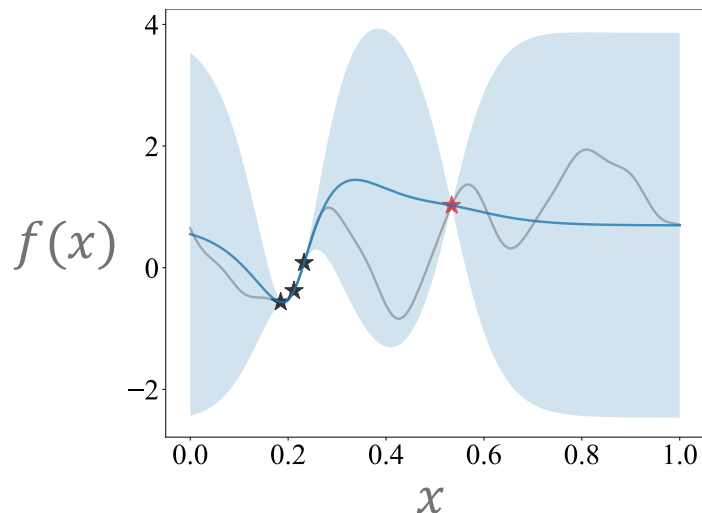
- *Closed*: open it
- *Opened*: **select & stop**

Discrete

Independent

Optimal policy: **Gittins index**

Cost-aware Bayesian Optimization

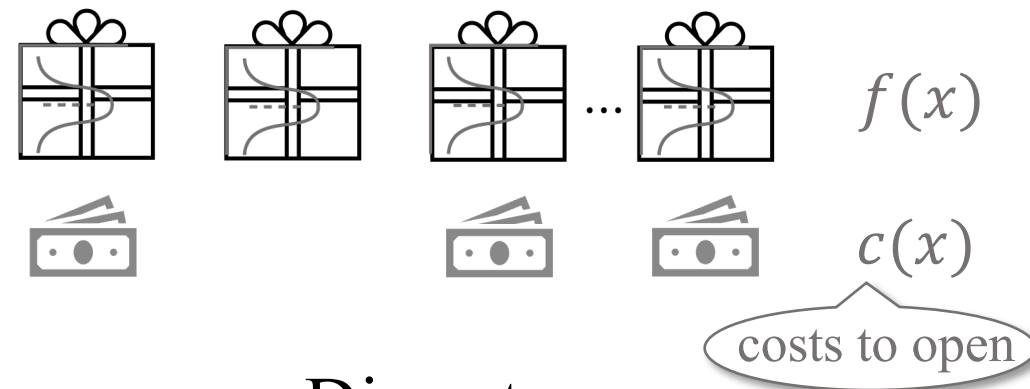


Continuous search space

Correlated function values

Pandora's Box

[Weitzman'79]



Discrete

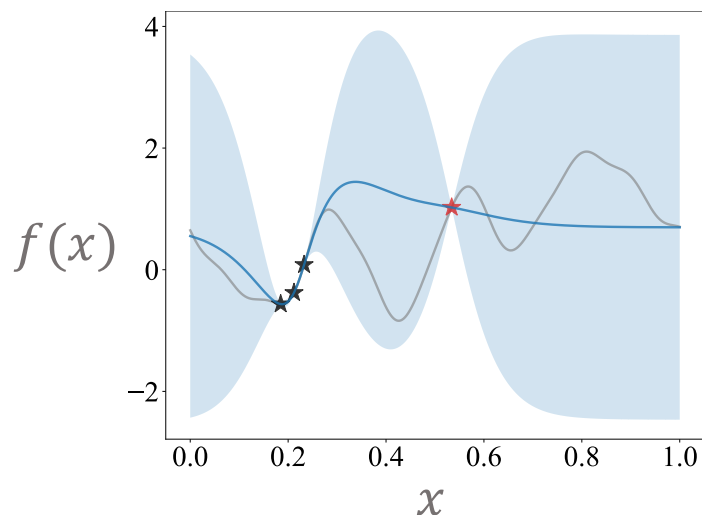
Independent

How to translate?



Optimal policy: Gittins index

Cost-aware Bayesian Optimization



Continuous search space

Correlated function values

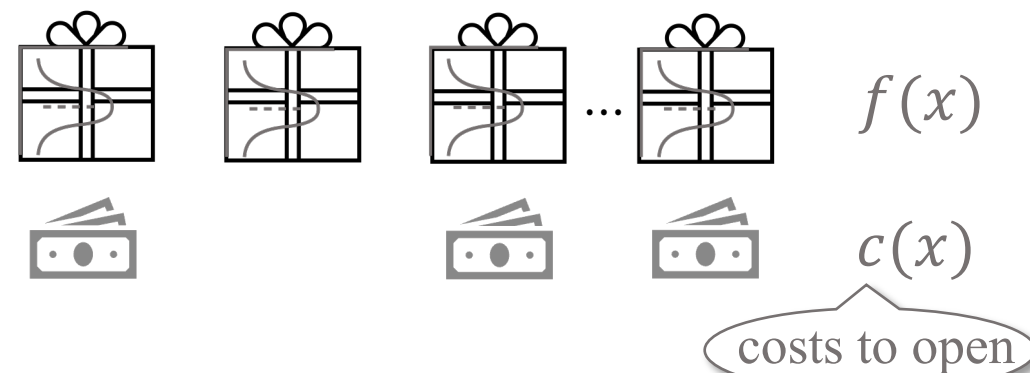
Our policy: $\text{GI}_{f|D,c}(x)$ $\xleftarrow[\text{take continuum limit}]{\text{incorporate posterior}}$ Optimal policy: $\text{GI}_{f,c}(x)$

Data

New!

Pandora's Box

[Weitzman'79]



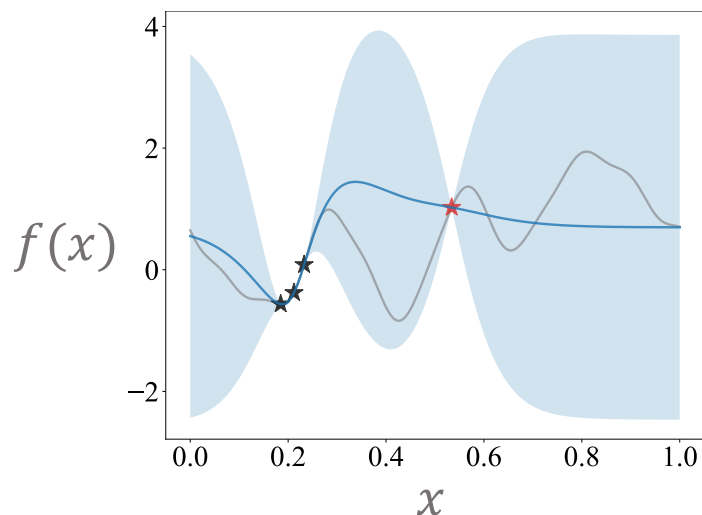
Discrete

Independent



"Cost-aware Bayesian Optimization via the Pandora's Box Gittins Index." NeurIPS'24.

Cost-aware Bayesian Optimization



Continuous search space

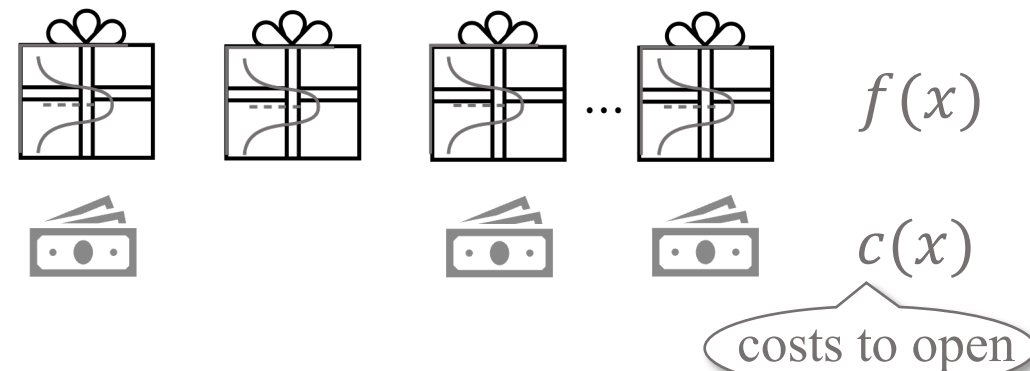
Correlated function values

Our policy: $\text{GI}_{f|D,c}(x)$

How to implement?

Pandora's Box

[Weitzman'79]



Discrete

Independent

Optimal policy: $\text{GI}_{f,c}(x)$

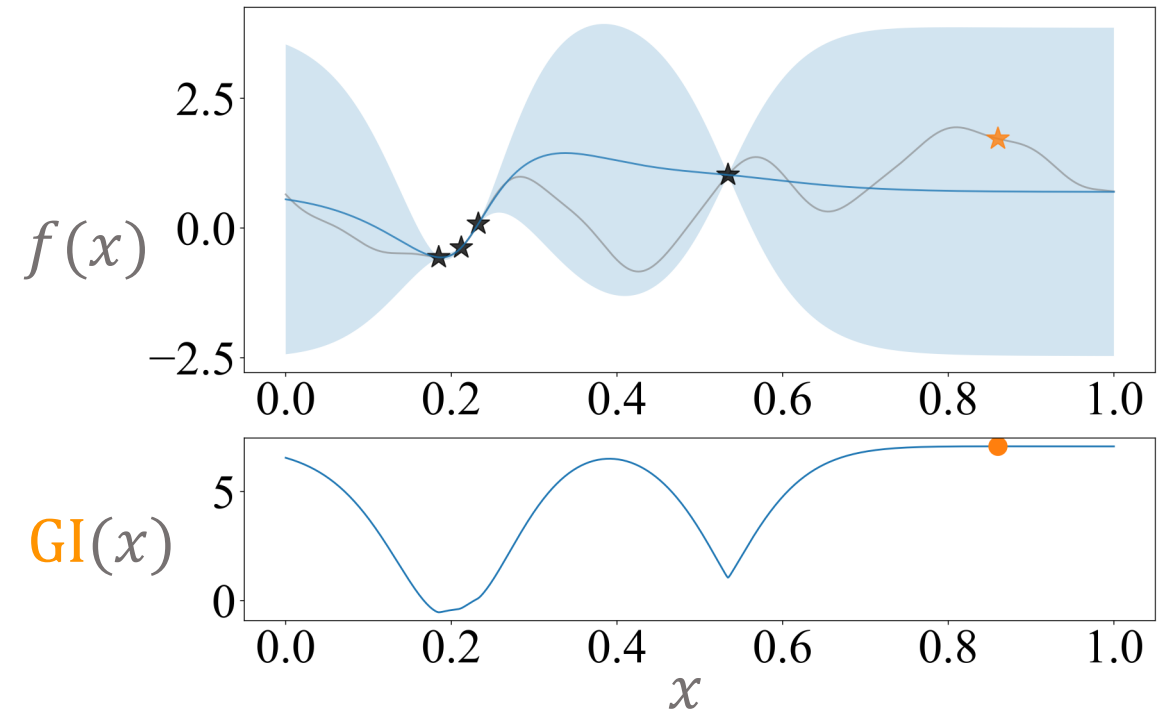
New!

incorporate posterior
take continuum limit

How to implement Gittins Indices?

$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.} \\ \mathbb{E}[\max(f(x) - g, 0) \mid D] = c(x)$$

Data



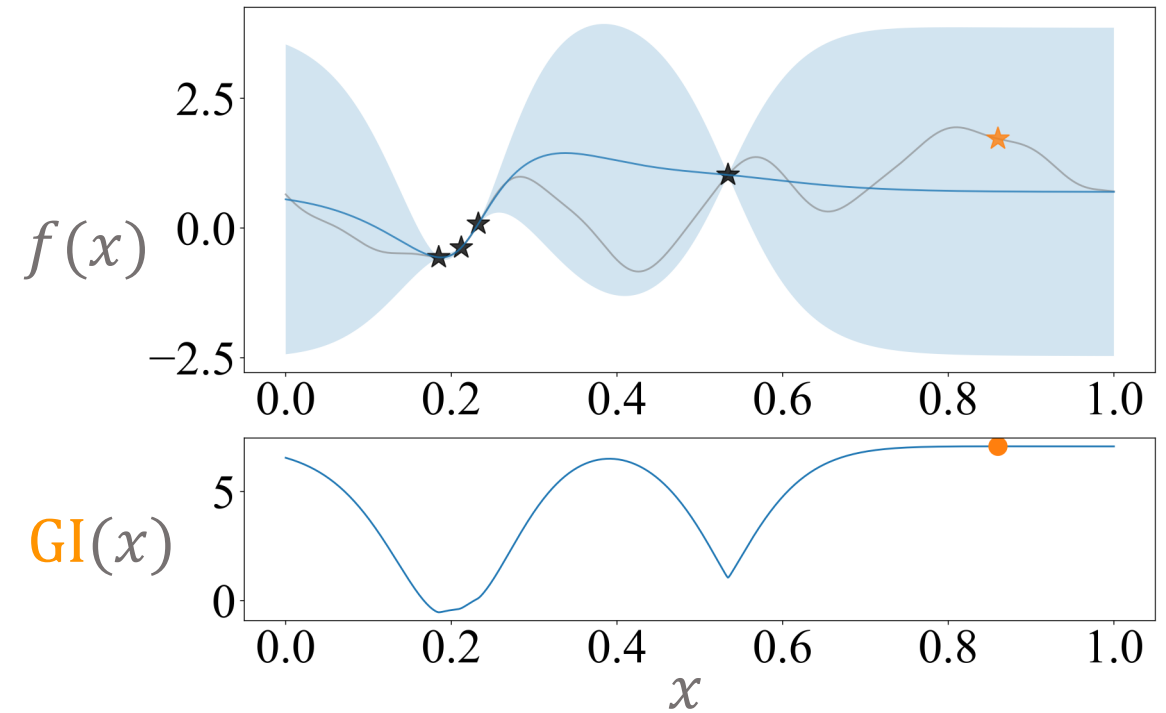
Gittins Indices

How to implement Gittins Indices?

$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.} \\ \mathbb{E}[\max(f(x) - g, 0) \mid D] = c(x)$$

analytical expression
& monotonicity in g

Data



Gittins Indices

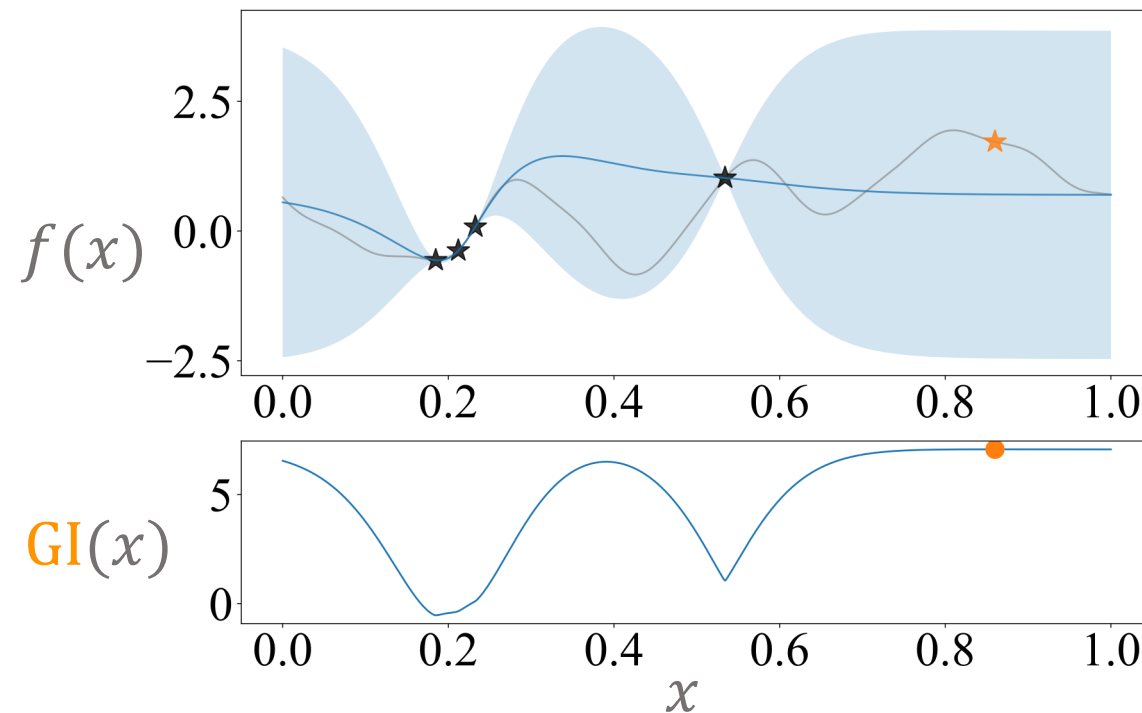
How to implement Gittins Indices?

$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.}$$
$$\mathbb{E}[\max(f(x) - g, 0) \mid D] = c(x)$$

analytical expression
& monotonicity in g

bisection search

Data



Gittins Indices

💡 Easy to compute!



"Cost-aware Bayesian Optimization via the Pandora's Box Gittins Index." NeurIPS'24.

How to implement Gittins Indices?

$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.} \\ \mathbb{E}[\max(f(x) - g, 0) \mid D] = c(x)$$

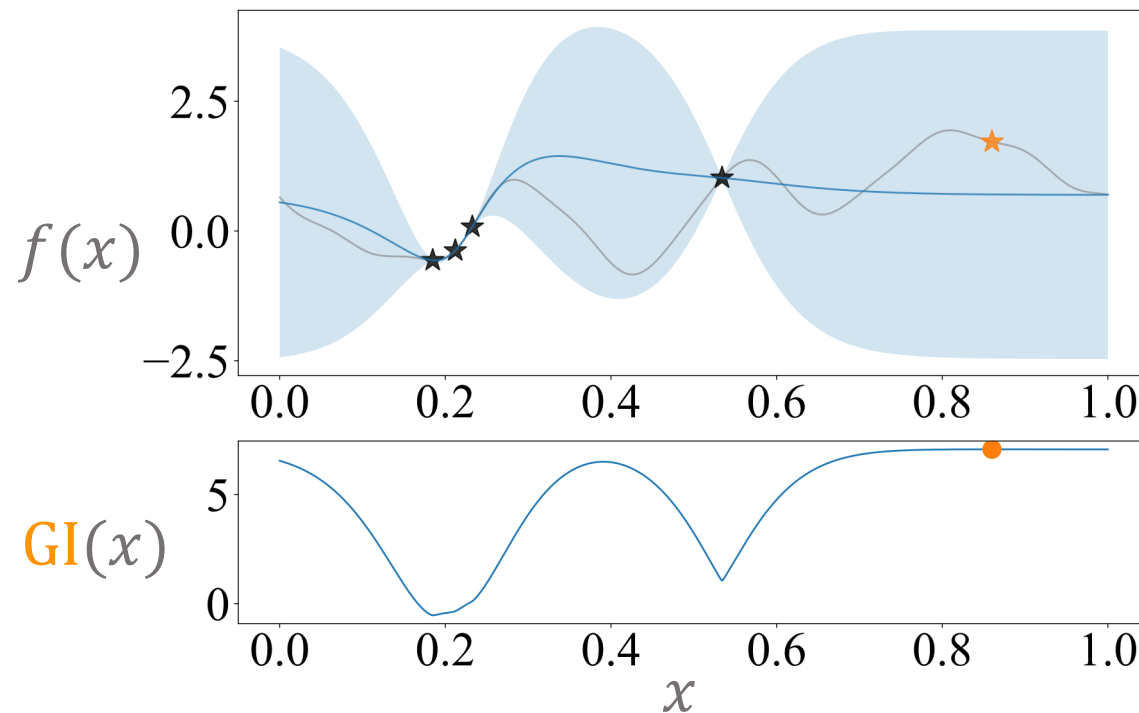
analytical expression
& monotonicity in g

bisection search

Data

$$x_{t+1} = \operatorname{argmax}_x \text{GI}_{f|D,c}(x)$$

Analytical gradient



Gittins Indices

💡 Easy to compute and optimize!



"Cost-aware Bayesian Optimization via the Pandora's Box Gittins Index." NeurIPS'24.

How to implement Gittins Indices?

$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.}$
 $\mathbb{E}[\max(f(x) - g, 0) \mid D] = c(x)$

bisection search

$\text{EI}_{f|D}(x; g)$

& monotonicity in g

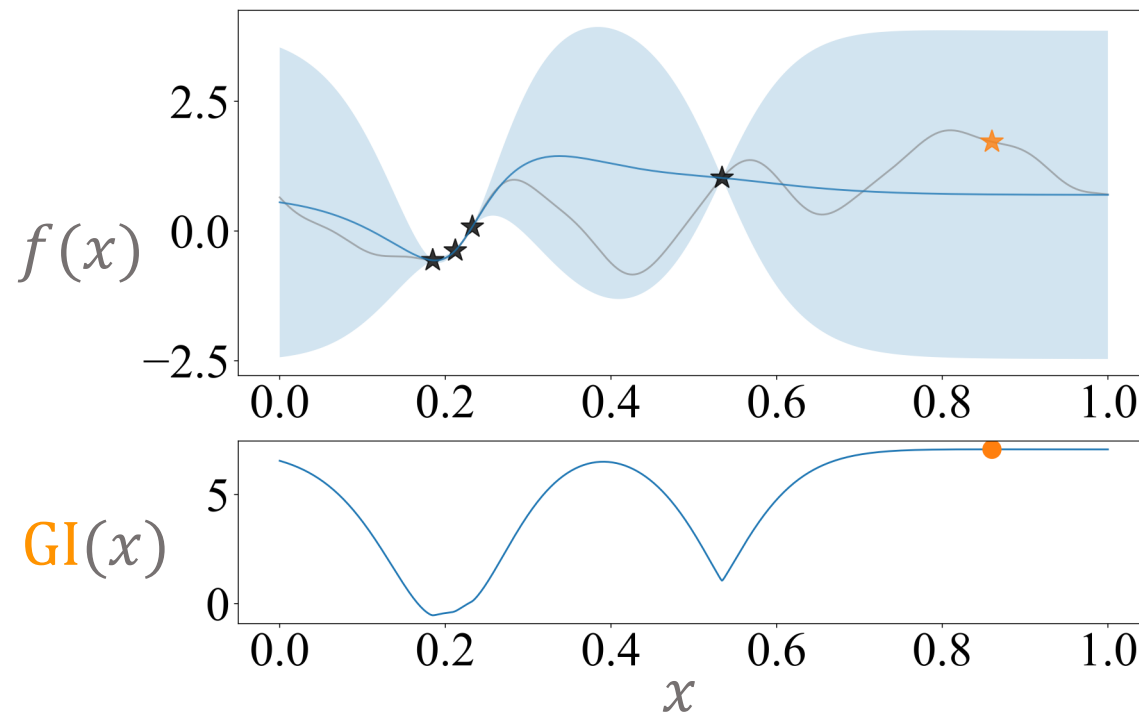
$$x_{t+1} = \operatorname{argmax}_x \text{GI}_{f|D,c}(x)$$

Analytical gradient

Data

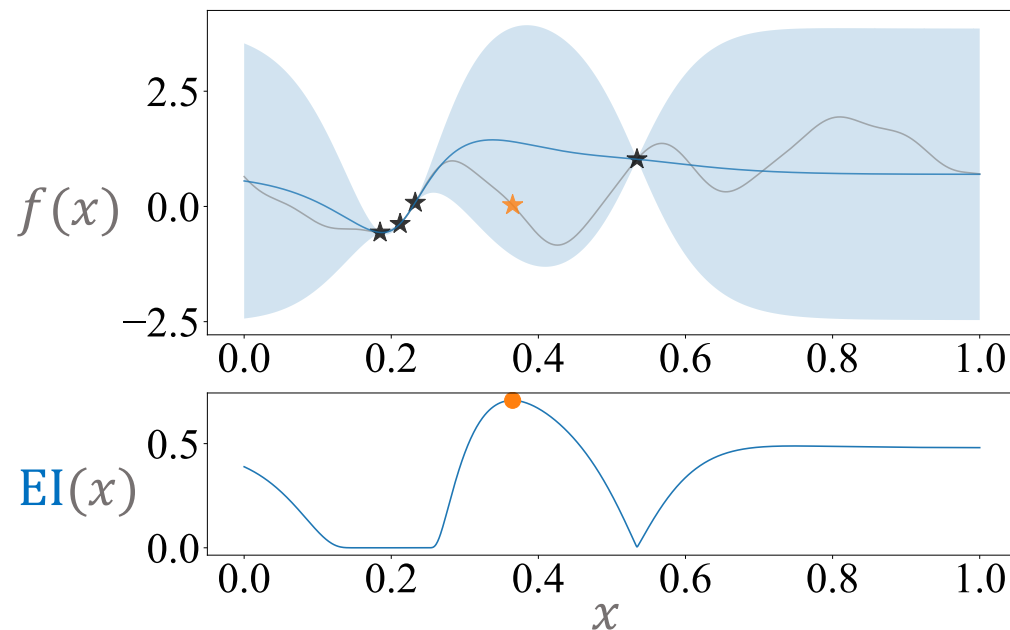


Easy to compute and optimize!



Gittins Indices

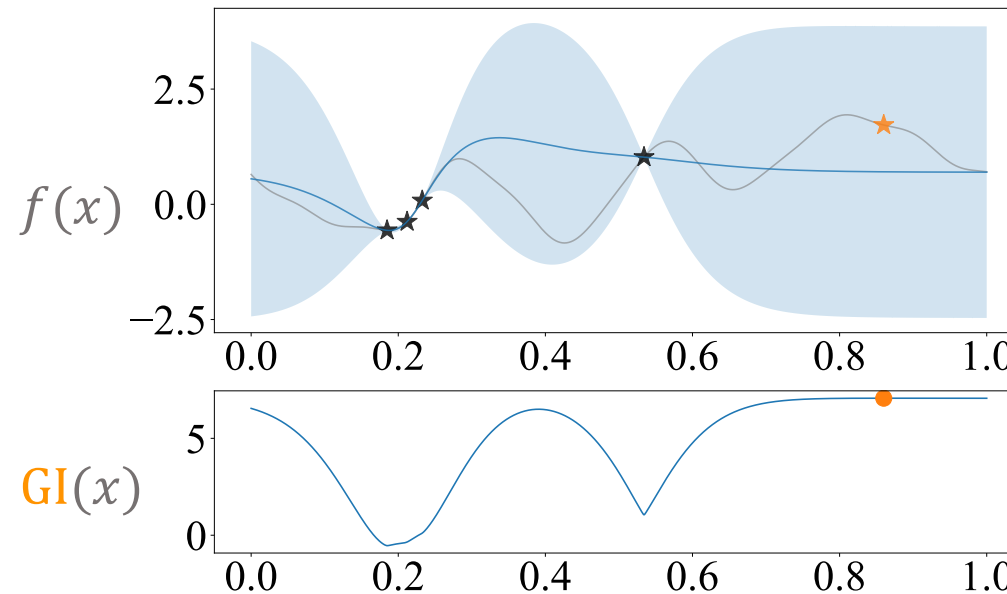
Expected Improvement



$$EI_{f|D}(x; y_{\text{best}}) := \mathbb{E}[\max(f(x) - y_{\text{best}}, 0) \mid D]$$

Acquisition rule: $\operatorname{argmax}_x EI_{f|D}(x; y_{\text{best}})$

Gittins Index

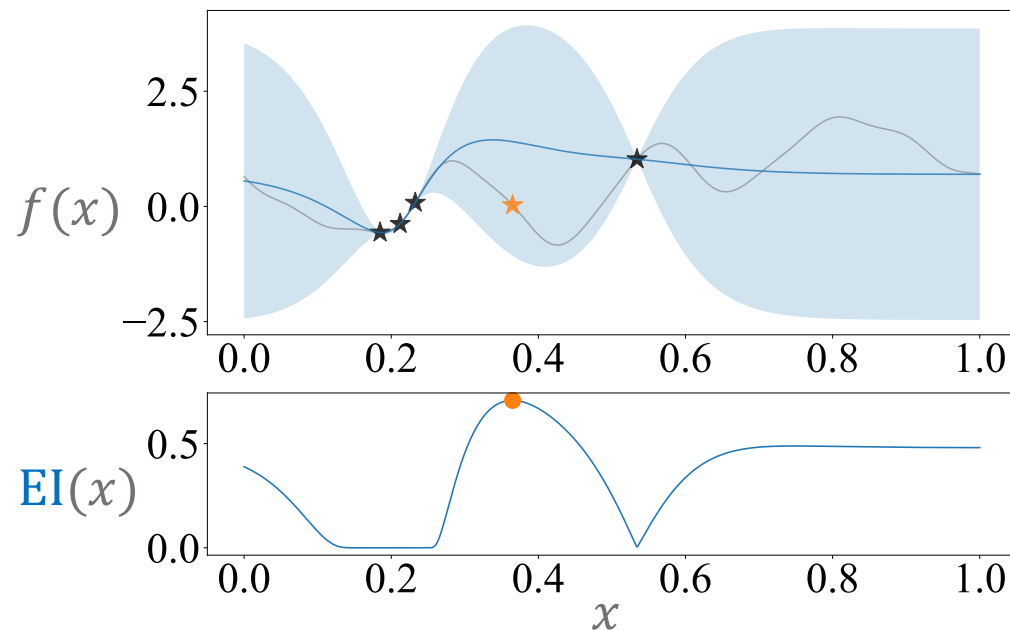


$$GI_{f|D,c}(x) := \text{solution } g \text{ s.t. } EI_{f|D}(x; g) = c(x)$$

where $EI_{f|D}(x; g) := \mathbb{E}[\max(f(x) - g, 0) \mid D]$

$$\operatorname{argmax}_x GI_{f|D,c}(x)$$

Expected Improvement

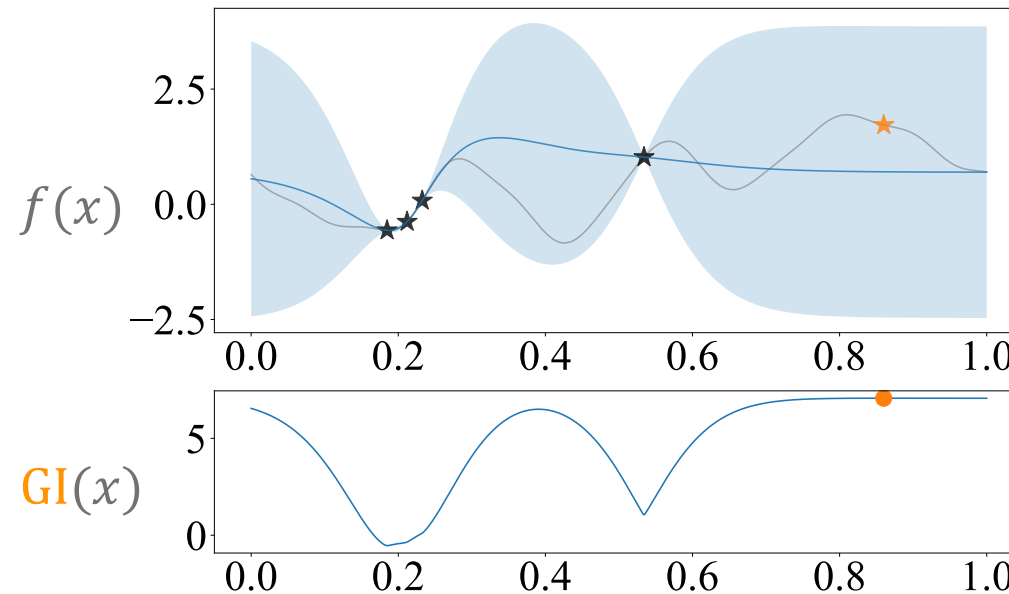


$$EI_{f|D}(x; \mathbf{y}_{\text{best}}) := \mathbb{E}[\max(f(x) - \mathbf{y}_{\text{best}}, 0) \mid D]$$

Acquisition rule: $\operatorname{argmax}_x EI_{f|D}(x; \mathbf{y}_{\text{best}})$

Stopping rule:

Gittins Index



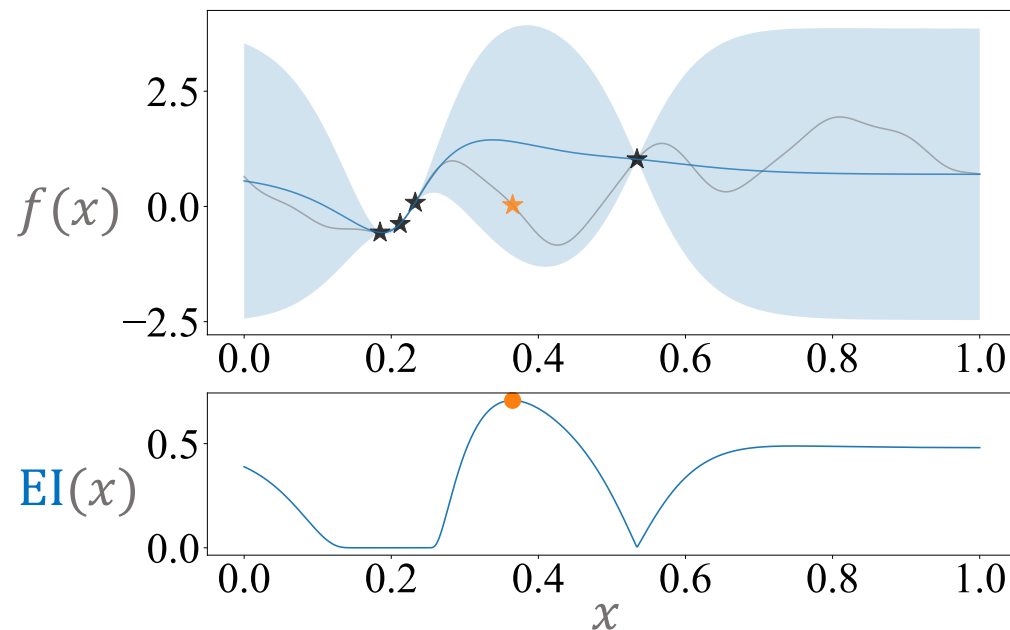
$$GI_{f|D,c}(x) := \text{solution } g \text{ s.t. } EI_{f|D}(x; g) = c(x)$$

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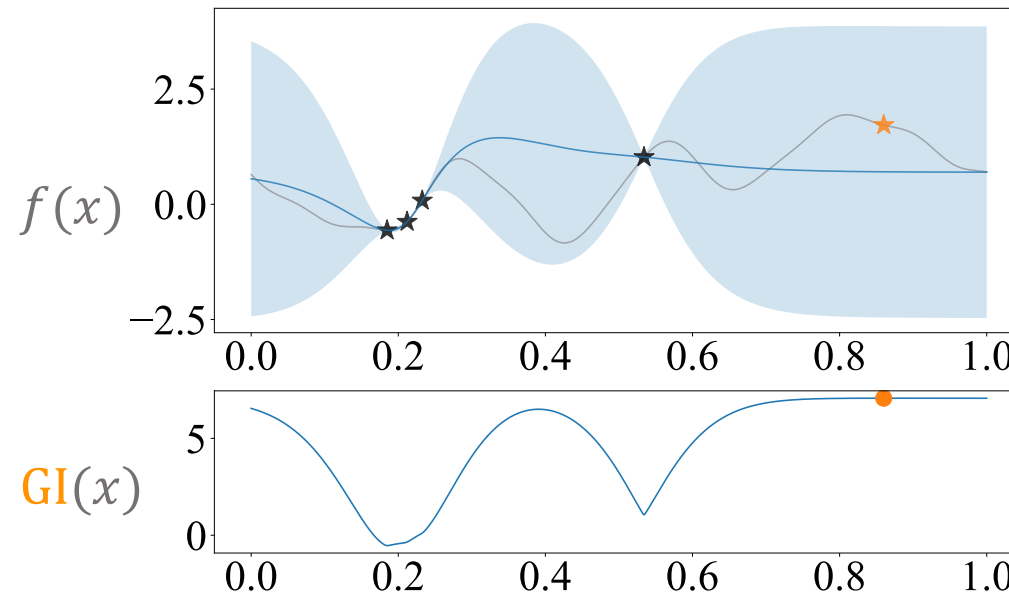
$$\operatorname{argmax}_x GI_{f|D,c}(x)$$

$$\tau: GI_{f|D,c}(x_\tau) \leq \mathbf{y}_{\text{best}}$$

Expected Improvement



Gittins Index



$$EI_{f|D}(x; \mathbf{y}_{\text{best}}) := \mathbb{E}[\max(f(x) - \mathbf{y}_{\text{best}}, 0) \mid D]$$

$$GI_{f|D,c}(x) := \text{solution } g \text{ s.t. } EI_{f|D}(x; g) = c(x) \\ \text{where } EI_{f|D}(x; g) := \mathbb{E}[\max(f(x) - g, 0) \mid D]$$

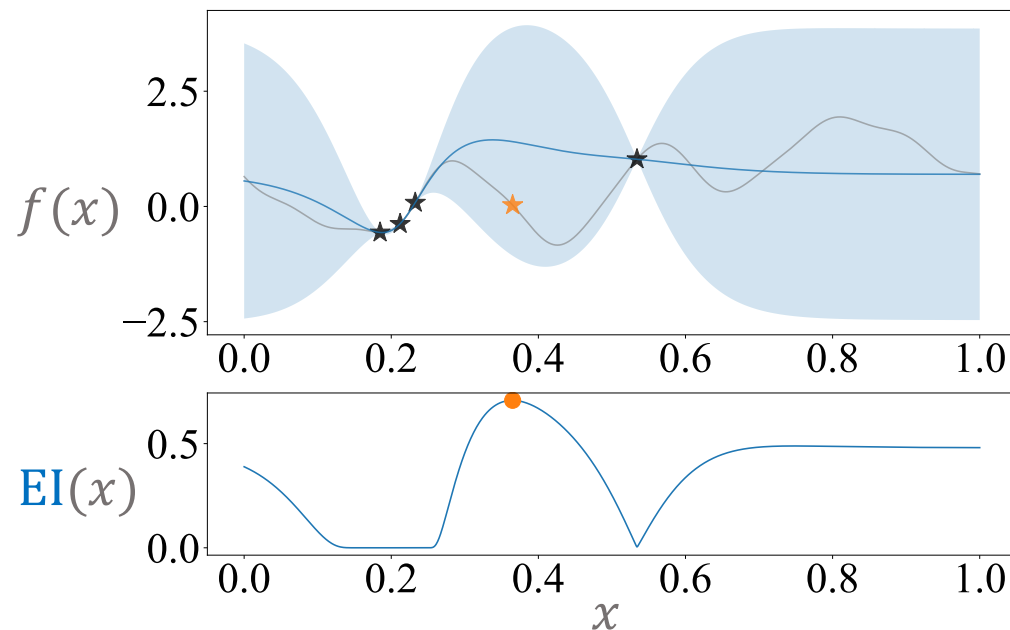
Acquisition rule: $\operatorname{argmax}_x EI_{f|D}(x; \mathbf{y}_{\text{best}})$

$\operatorname{argmax}_x GI_{f|D,c}(x)$

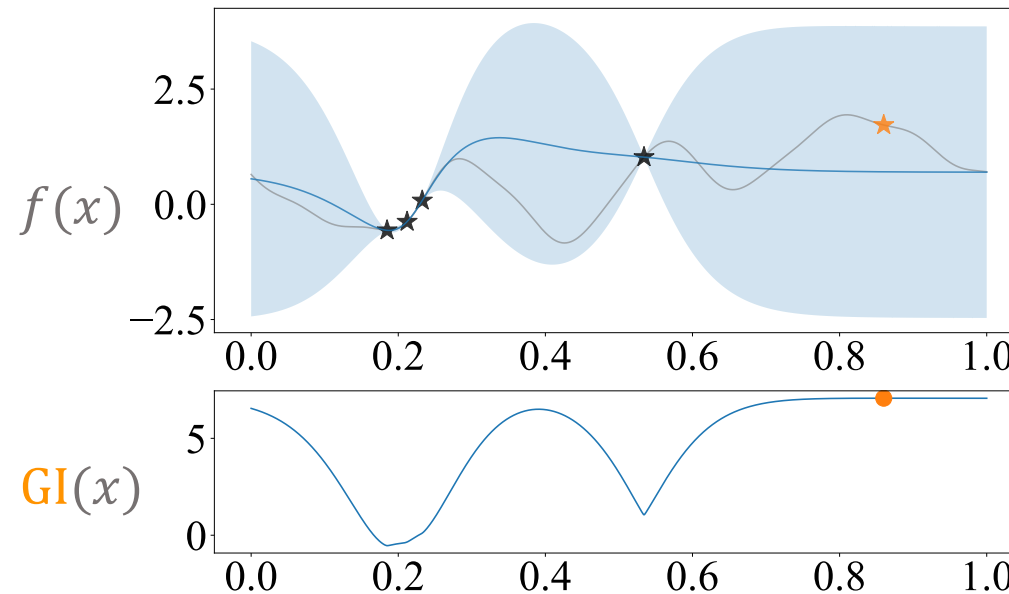
Stopping rule:

$$\tau: GI_{f|D,c}(x_\tau) \leq \mathbf{y}_{\text{best}} \\ \Leftrightarrow \tau: EI_{f|D}(x; \mathbf{y}_{\text{best}}) \leq EI_{f|D}\left(x; GI_{f|D,c}(x_\tau)\right) = c(x_\tau)$$

Expected Improvement



Gittins Index



$$\text{EI}_{f|D}(x; \mathbf{y}_{\text{best}}) := \mathbb{E}[\max(f(x) - \mathbf{y}_{\text{best}}, 0) \mid D]$$

$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t. } \text{EI}_{f|D}(x; g) = c(x)$$

where $\text{EI}_{f|D}(x; g) := \mathbb{E}[\max(f(x) - g, 0) \mid D]$

Acquisition rule: $\arg\max_x \text{EI}_{f|D}(x; \mathbf{y}_{\text{best}})$

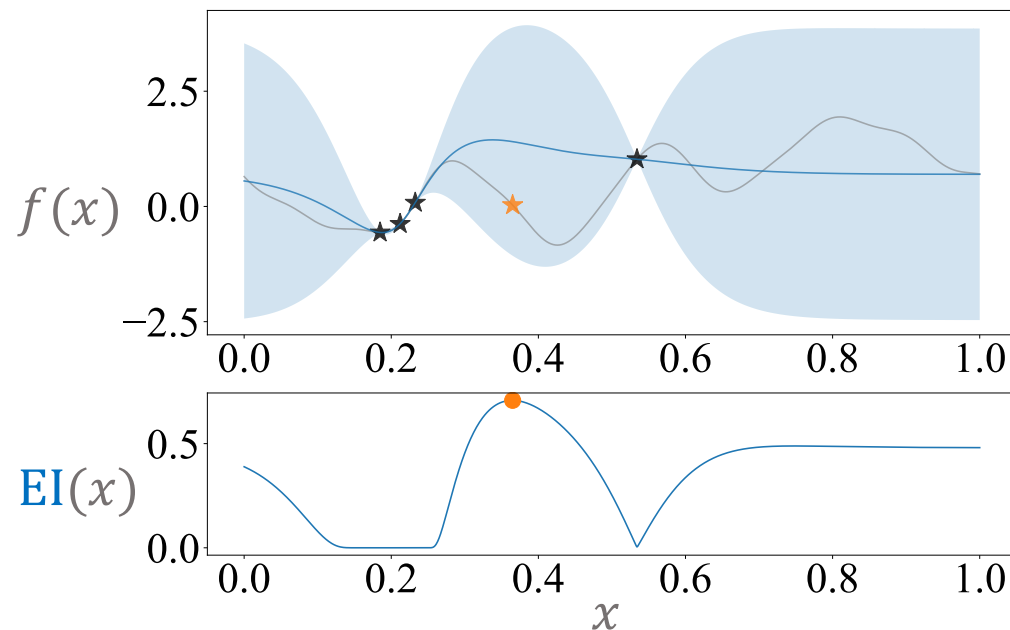
$\arg\max_x \text{GI}_{f|D,c}(x)$

Stopping rule: $\tau: \text{EI}_{f|D}(x_\tau; \mathbf{y}_{\text{best}}) \leq c(x_\tau)$

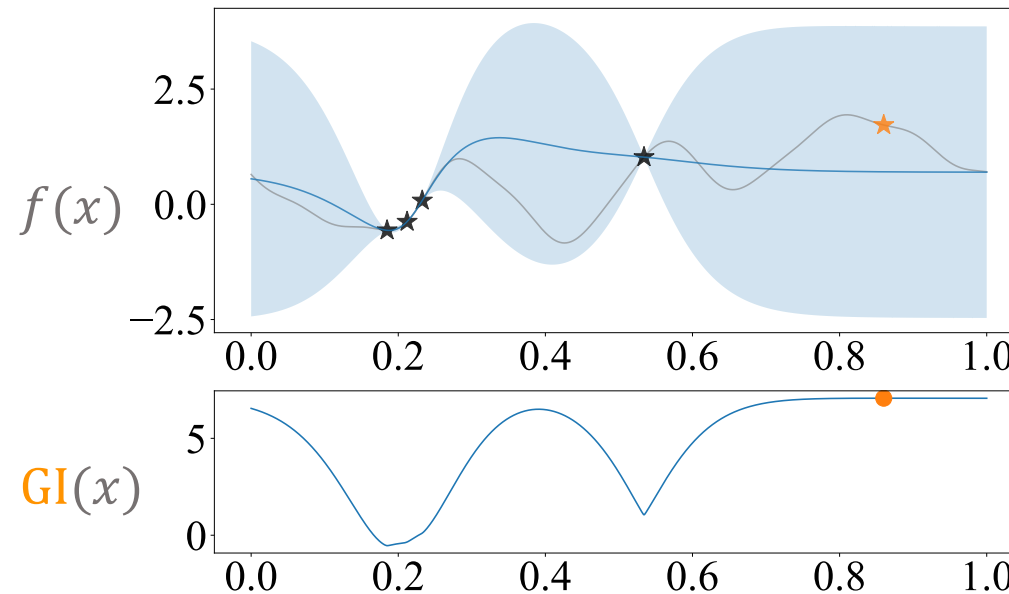
$\tau: \text{GI}_{f|D,c}(x_\tau) \leq \mathbf{y}_{\text{best}}$

$$\Leftrightarrow \tau: \text{EI}_{f|D}(x; \mathbf{y}_{\text{best}}) \leq \text{EI}_{f|D}\left(x; \text{GI}_{f|D,c}(x_\tau)\right) = c(x_\tau)$$

Expected Improvement



Gittins Index



$$\text{EI}_{f|D}(x; \mathbf{y}_{\text{best}}) := \mathbb{E}[\max(f(x) - \mathbf{y}_{\text{best}}, 0) \mid D]$$

$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t. } \text{EI}_{f|D}(x; g) = c(x)$$

where $\text{EI}_{f|D}(x; g) := \mathbb{E}[\max(f(x) - g, 0) \mid D]$

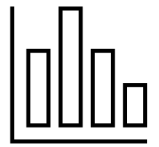
Acquisition rule: $\arg\max_x \text{EI}_{f|D}(x; \mathbf{y}_{\text{best}})$

$\arg\max_x \text{GI}_{f|D,c}(x)$

Stopping rule: $\tau: \text{EI}_{f|D}(x_\tau; \mathbf{y}_{\text{best}}) \leq c(x_\tau)$
 $\Leftrightarrow \tau: \text{EIPC}_{f|D,c}(x_\tau; \mathbf{y}_{\text{best}}) \leq 1$

$\tau: \text{GI}_{f|D,c}(x_\tau) \leq \mathbf{y}_{\text{best}}$
 $\Leftrightarrow \tau: \text{EI}_{f|D}(x; \mathbf{y}_{\text{best}}) \leq \text{EI}_{f|D}(x; \text{GI}_{f|D,c}(x_\tau)) = c(x_\tau)$

Expected Improvement vs Gittins Index



Varying evaluation costs

$$\text{EIPC}_c(x; y_{\text{best}}) = \text{EI}(x; y_{\text{best}})/c(x) \quad \text{GI}_c(x) := \text{solution } g \text{ s.t. } \text{EI}(x; g) = c(x)$$

Arbitrarily bad

Costs built-in



Smart stopping time

$$\tau: \text{EI}(x_\tau; y_{\text{best}}) \leq \theta$$

Which threshold?

$$\tau: \text{GI}_c(x_\tau) \leq y_{\text{best}} \\ \Leftrightarrow \tau: \text{EIPC}_c(x_\tau; y_{\text{best}}) \leq 1$$

Derived shared stopping rule

Temporal (one-step) simplification to MDP

Spatial simplification to MDP



"Cost-aware Bayesian Optimization via the Pandora's Box Gittins Index." NeurIPS'24.



"Cost-aware Stopping for Bayesian Optimization." ICML'26

Theoretical Guarantee and Empirical Validation

Theorem (Safeguard Guarantee)

$$\mathbb{E}[R(\text{ours}; \text{PBGI})] \leq R[\text{stopping immediately}]$$

or LogEIPC

cost-adjusted regret

Implication:

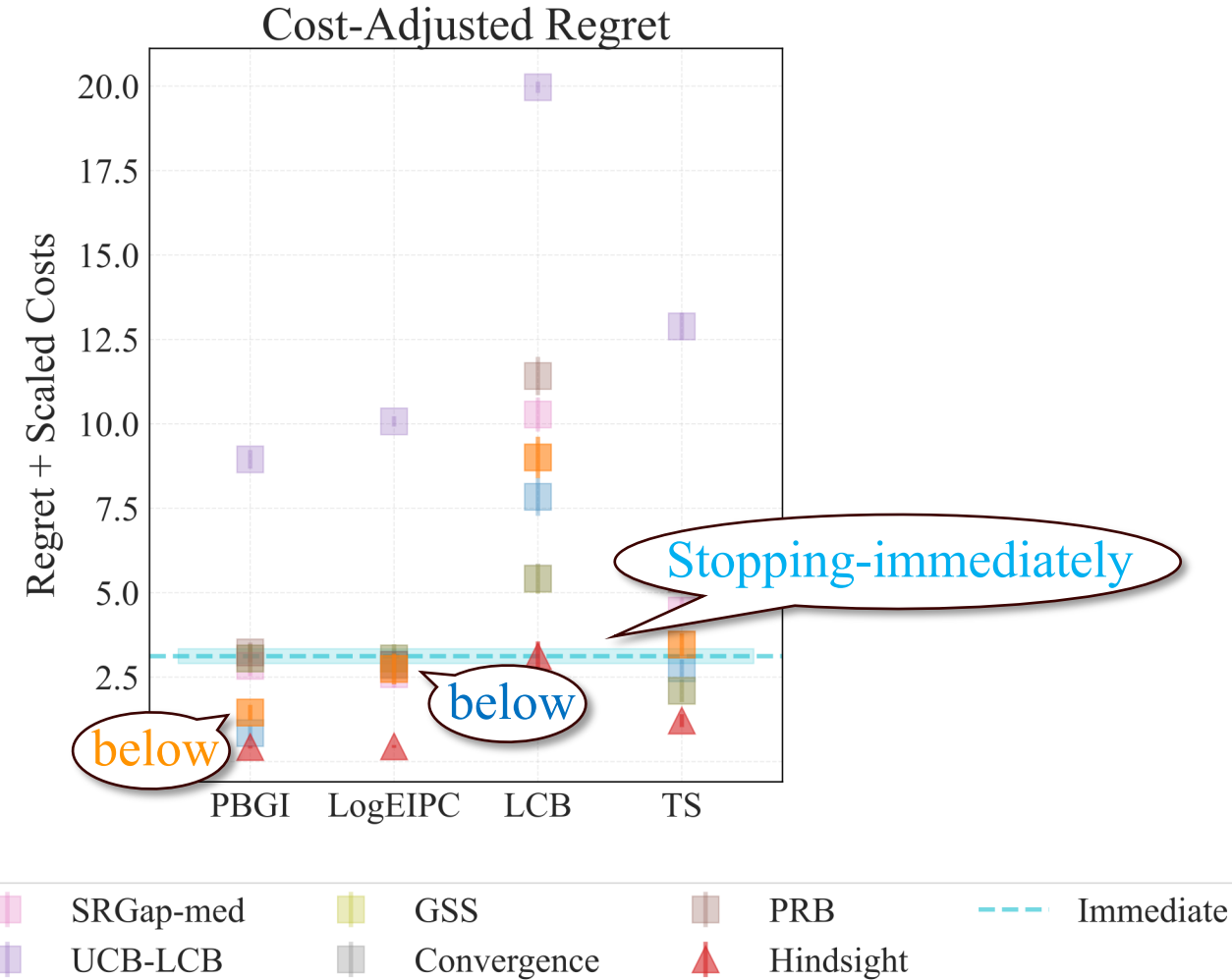
- Matches the **best achievable performance in the worst case** (evaluations are all very costly).

- Avoids over-spending** — a property many cost-unaware stopping rules lack.

New

Proof idea: For all $t < \tau$, $\text{EI}(x_{t+1}) \geq c(x_{t+1})$.

stopping time

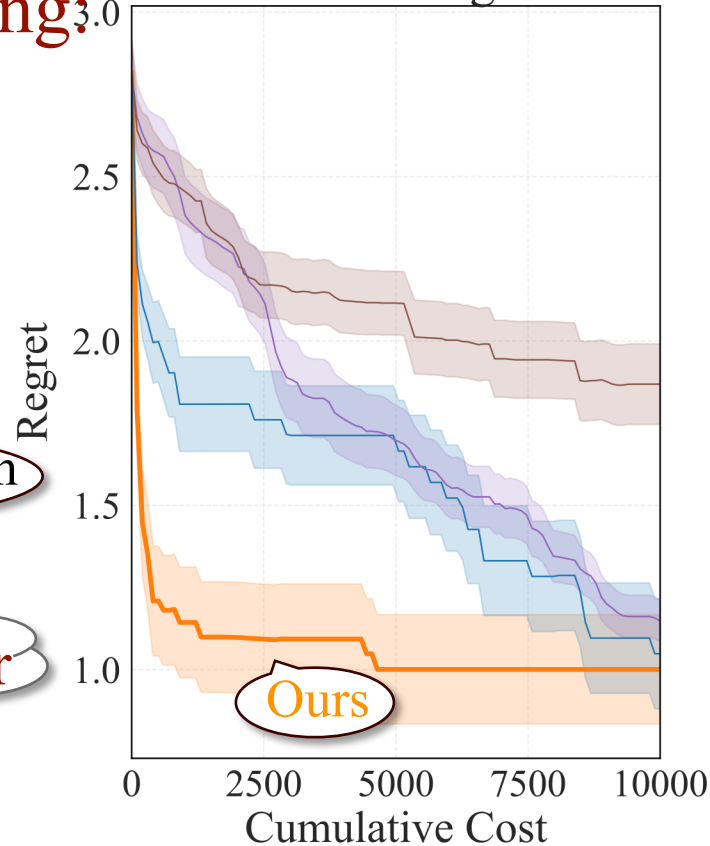


"Cost-aware Stopping for Bayesian Optimization." ICML'26

Gittins Index vs Baselines on LCBench

ML model training

Fixed-budget

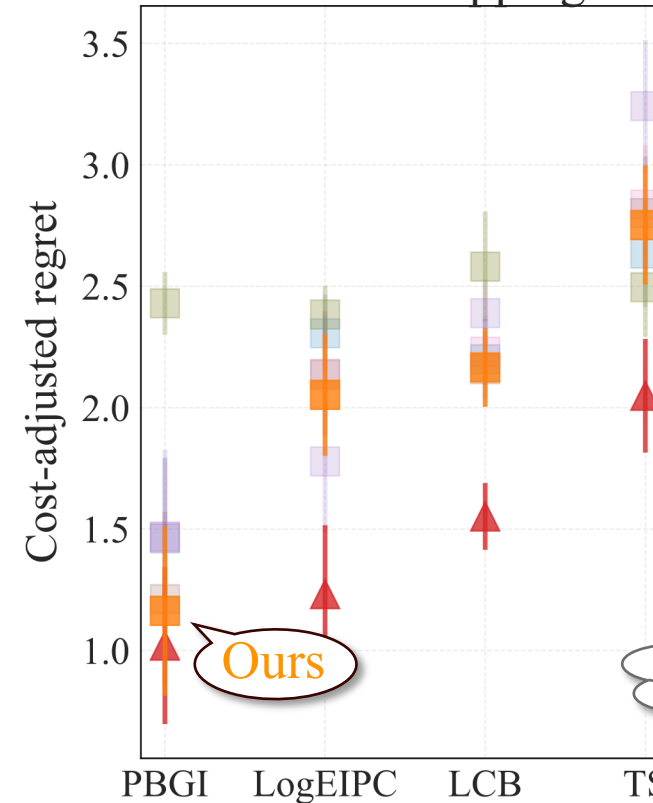


gap to the optimum

Lower the better

Ours

Flexible-stopping



Lower the better

Cost-aware

Cost-unaware



Acquisition rules

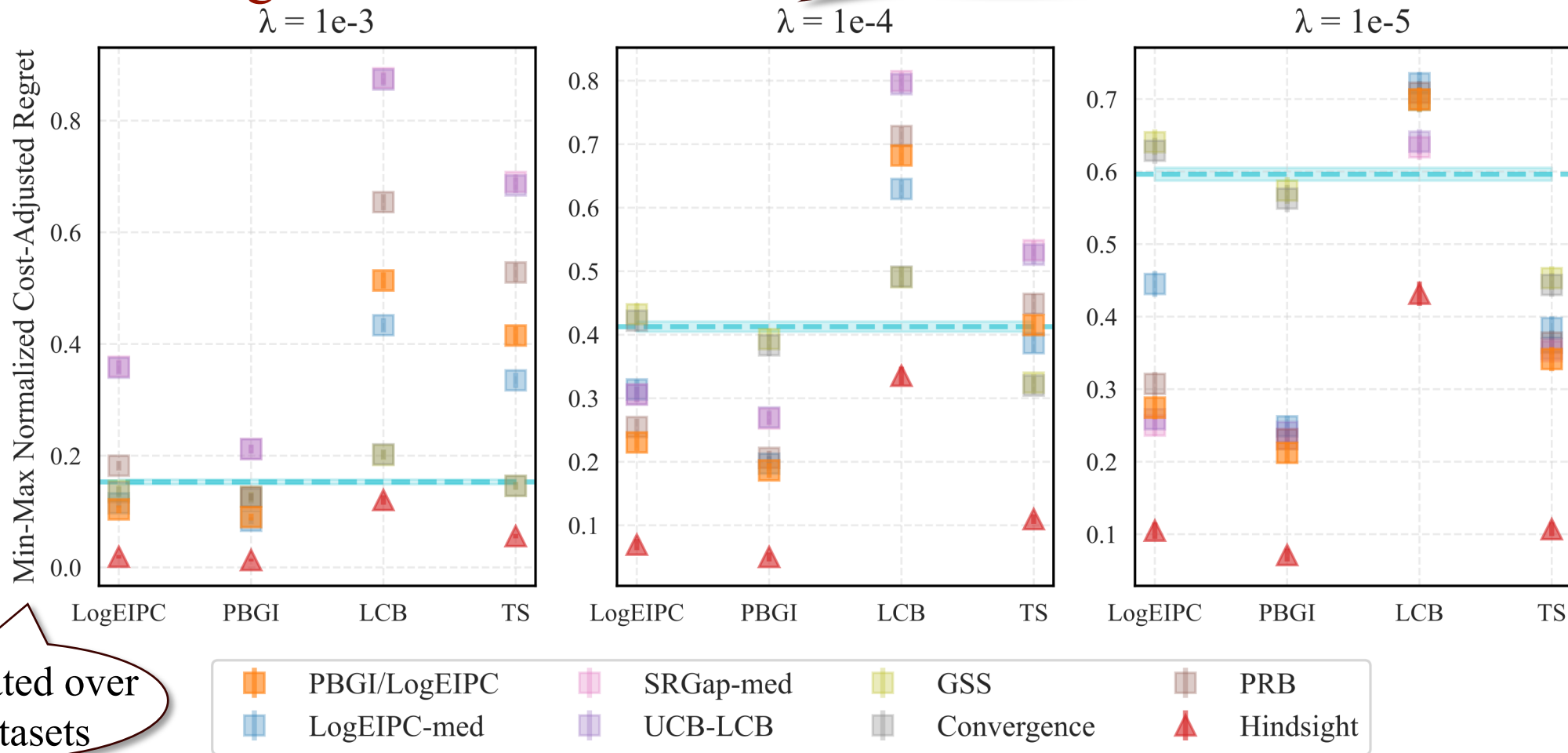
Stopping rules

Not a real baseline

Gittins Index vs Baselines on LCBench

ML model training:

Cost-scaling factor



"Cost-aware Stopping for Bayesian Optimization." ICML'26

Gittins Index vs Baselines on LCBench

ML model training:

λ	Acq	Top 1		Top 2		Top 3	
		Large	All	Large	All	Large	All
10^{-3}	PBGI	20.0%	20.0%	70.0%	48.6%	80.0%	60.0%
	LogEIPC	40.0%	31.4%	65.0%	45.7%	70.0%	62.8%
10^{-4}	PBGI	40.0%	28.6%	70.0%	51.5%	80.0%	62.9%
	LogEIPC	30.0%	22.9%	50.0%	45.8%	85.0%	68.7%
10^{-5}	PBGI	50.0%	28.6%	70.0%	51.5%	90.0%	74.4%
	LogEIPC	20.0%	20.0%	50.0%	48.6%	85.0%	71.5%

(All: 35 datasets, Large: 20 datasets w/ $>10,000$ instances)

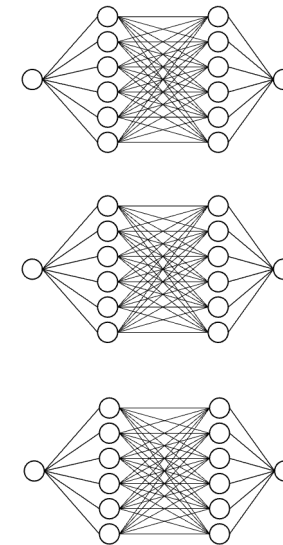


"Cost-aware Stopping for Bayesian Optimization." ICML'26

Part II:

Towards Bayesian Optimization with Multi-Stage Feedback (Bandit-like Bayesian Decision Problems) via Markov Chain Gittins Indices

Model	Q1	Q2	Q3	...	avg performance
 ChatGPT GPT-4o	✓	?	?	...	?
 Claude 3.5 Sonnet	?	✓	?	...	?
 Deepseek	✓	? Y_{ij}	?	...	?
 Gemini 1.5 Pro	✗	?	✓	...	?
 Llama 3.1-70B	?	✗	?	...	?
 Mistral Large	?	?	✗	...	?
...	...	...	...	...	...



Best-Model Identification in LLM Evaluation

LLM evaluation:

Model



Model	Q1	Q2	Q3	...	avg performance
 ChatGPT GPT-4o	✓	?	?	...	?
 Claude 3.5 Sonnet	?	✓	?	...	?
 Deepseek	✓	$? Y_{ij}$	?	...	?
 Gemini 1.5 Pro	✗	?	✓	...	?
 Llama 3.1-70B	?	✗	?	...	?
 Mistral Large	?	?	✗	...	?
...	...	...	...	...	...



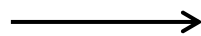
Average score
over questions

Goal: find LLM with best average score
(find row with best latent mean for a matrix)

Best-Model Identification in LLM Evaluation

LLM evaluation:

Model



Model	Q1	Q2	Q3	...	avg performance
 ChatGPT GPT-4o	✓	?	?	...	?
 Claude 3.5 Sonnet	?	✓	?	...	?
 Deepseek	✓	$? Y_{ij}$	?	...	?
 Gemini 1.5 Pro	✗	?	✓	...	?
 Llama 3.1-70B	?	✗	?	...	?
 Mistral Large	?	?	✗	...	?
...	...	...	...	...	...

\$



Average score
over questions

Goal: find LLM with best average score
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Best-Model Identification in LLM Evaluation

LLM evaluation:

Model



Model	Q1	Q2	Q3	...	avg performance
 ChatGPT GPT-4o	✓	?	?	...	?
 Claude 3.5 Sonnet	?	✓	?	...	?
 Deepseek	✓	? y_{ij}	?	...	?
 Gemini 1.5 Pro	✗	?	✓	...	?
 Llama 3.1-70B	?	✗	?	...	?
 Mistral Large	?	?	✗	...	?
...	...	...	...	...	...

\$\$\$



Average score
over questions

Goal: find LLM with best average score
(find row with best latent mean for a matrix)

Best-Model Identification in LLM Evaluation

LLM evaluation:

Model



Model	Q1	Q2	Q3	...	avg performance
 ChatGPT GPT-4o	✓	?	?	...	?
 Claude 3.5 Sonnet	?	✓	?	...	?
 Deepseek	✓	? Y_{ij}	?	...	?
 Gemini 1.5 Pro	✗	?	✓	...	?
 Llama 3.1-70B	?	✗	?	...	?
 Mistral Large	?	?	✗	...	?
...	...	...	...	...	...

\$\$\$



Average score over questions

Goal: find LLM with best average score
(find row with best **latent** mean for a matrix)

Cost: full evaluation



Reward:
average score

Pandora's Box



...



Best-Model Identification in LLM Evaluation

LLM evaluation:

Model



Model	Q1	Q2	Q3	...	avg performance
ChatGPT GPT-4o	✓	?	?	...	?
Claude 3.5 Sonnet	?	✓	?	...	?
Deepseek	✓	$? Y_{ij}$	?	...	?
Gemini 1.5 Pro	✗	?	✓	...	?
Llama 3.1-70B	?	✗	?	...	?
Mistral Large	?	?	✗	...	?
...	...	...	...	...	...

\$



Average score
over questions

Goal: find LLM with best average score
(find row with best latent mean for a matrix)

Best-Model Identification in LLM Evaluation

LLM evaluation:

Model



Model	Q1	Q2	Q3	...	avg performance
 ChatGPT GPT-4o	✓	?	?	...	?
 Claude 3.5 Sonnet	?	✓	?	...	?
 Deepseek	✓	$? Y_{ij}$	?	...	?
 Gemini 1.5 Pro	✗	?	✓	...	?
 Llama 3.1-70B	?	✗	?	...	?
 Mistral Large	?	?	✗	...	?
...	...	...	...	...	...

\$\$\$



Average score
over questions

Goal: find LLM with best average score
(find row with best latent mean for a matrix)

Best-Model Identification in LLM Evaluation

LLM evaluation:

Model



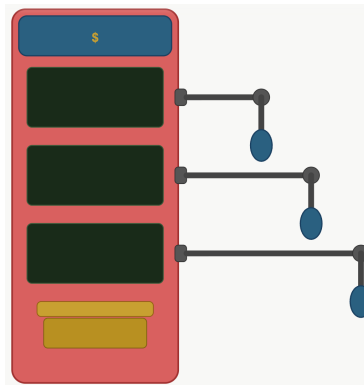
Model	Q1	Q2	Q3	...	avg performance
ChatGPT GPT-4o	✓	?	?	...	?
Claude 3.5 Sonnet	?	✓	?	...	?
Deepseek	✓	? y_{ij}	?	...	?
Gemini 1.5 Pro	✗	?	✓	...	?
Llama 3.1-70B	?	✗	?	...	?
Mistral Large	?	?	✗	...	?
...	...	...	...	...	...

\$\$\$



Average score over questions

Goal: find LLM with best average score
(find row with best latent mean for a matrix)

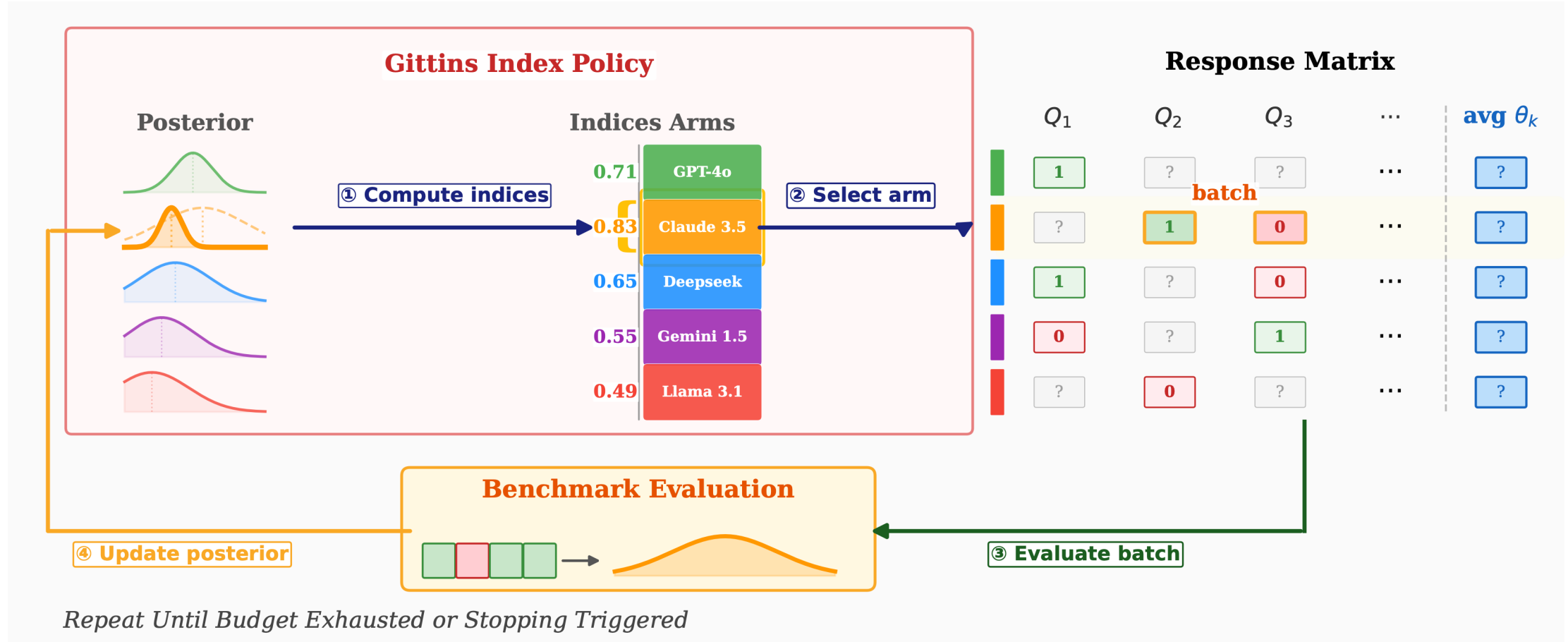


Cost-unaware & fixed-iteration

BanditEval
(Zhou et al. ICLR'25)

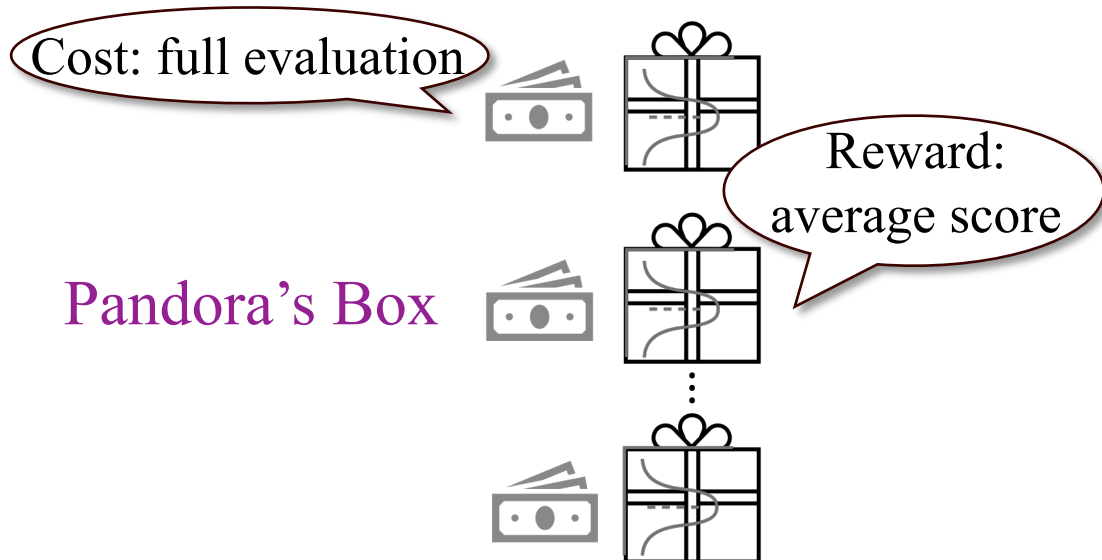
Best-Model Identification in LLM Evaluation

LLM evaluation:

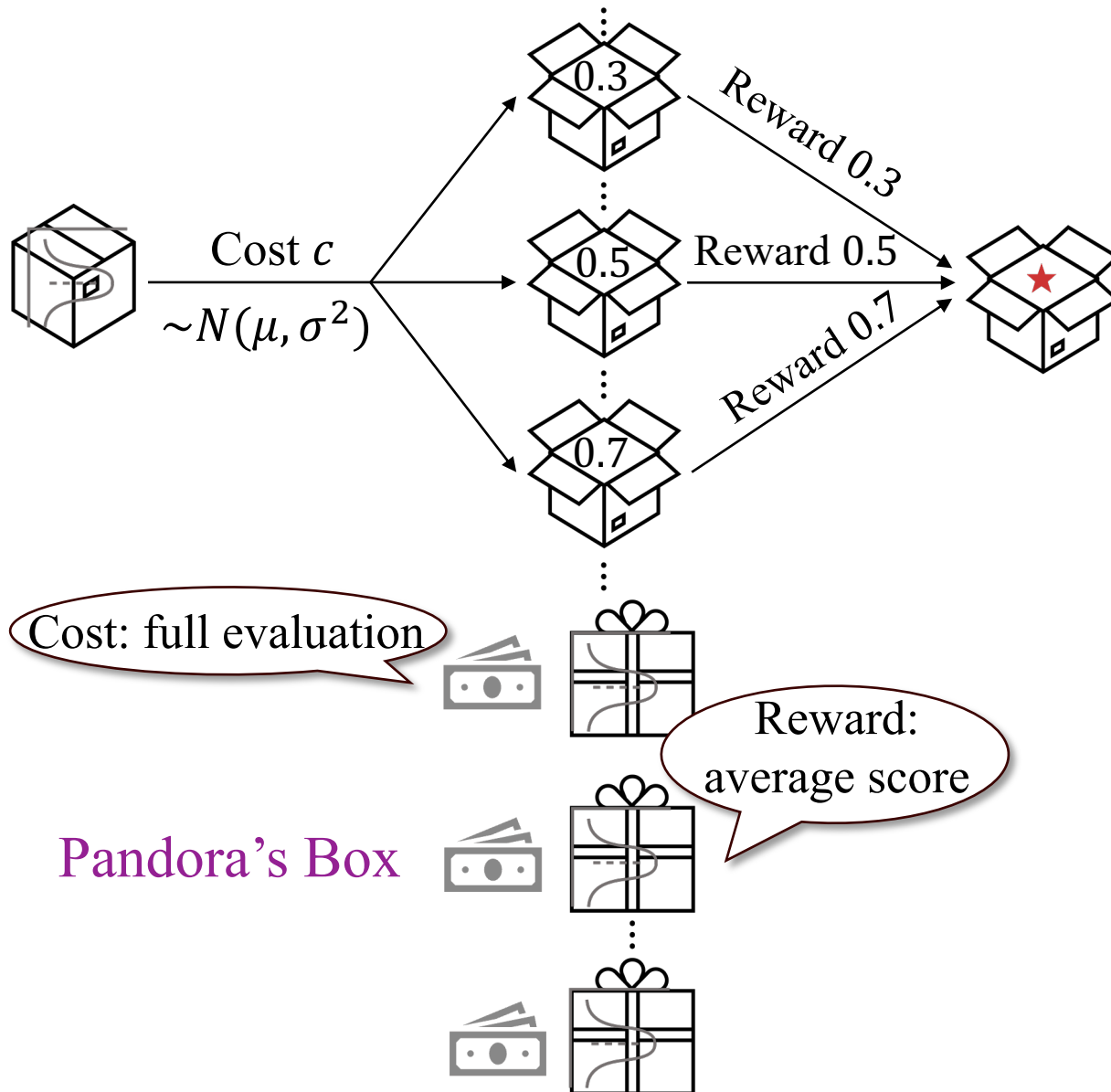


"Efficient Cost-Aware LLM Evaluation via Bayesian Bandit Gittins Indices." ICML'26 Workshop DEMO.

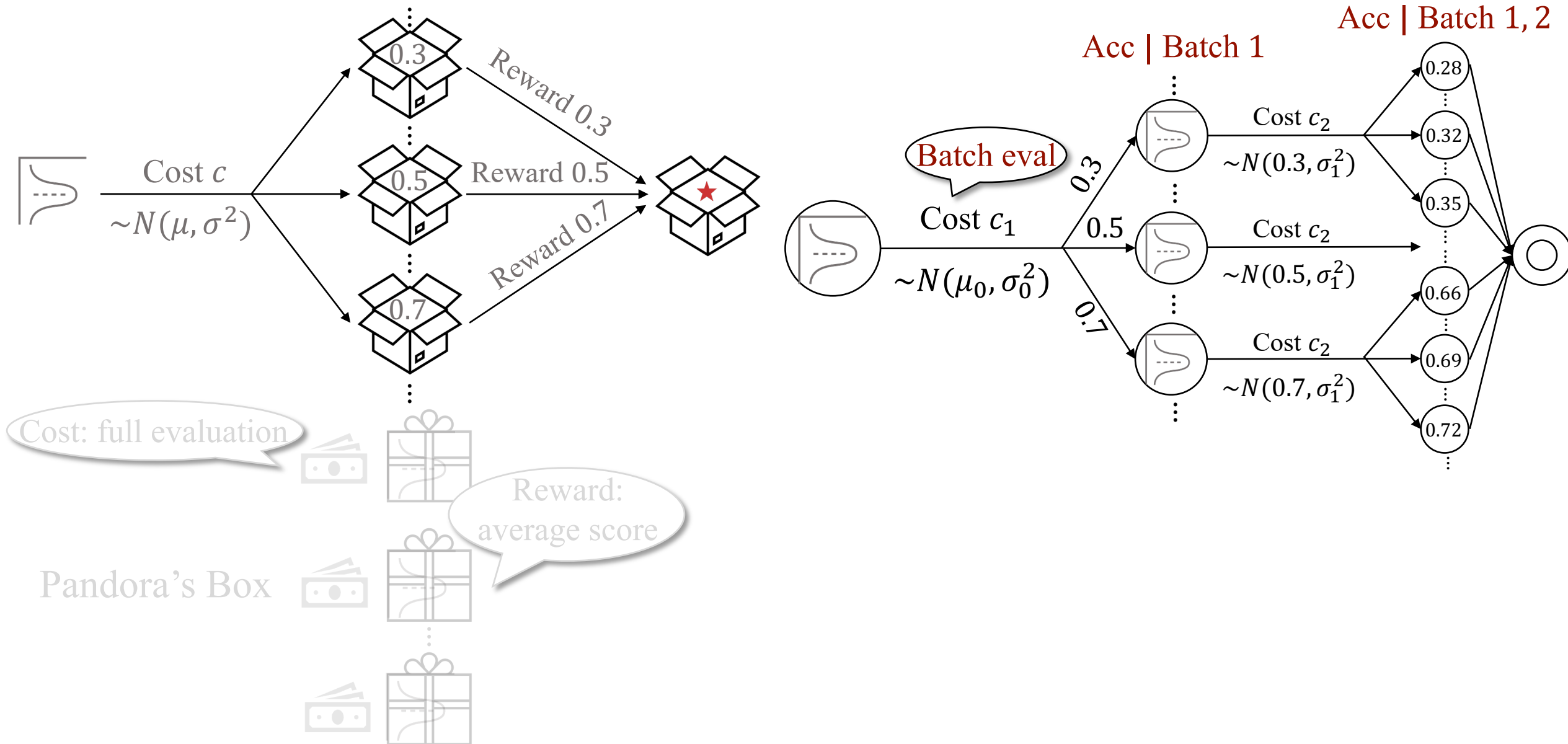
How to update posterior belief of latent mean?



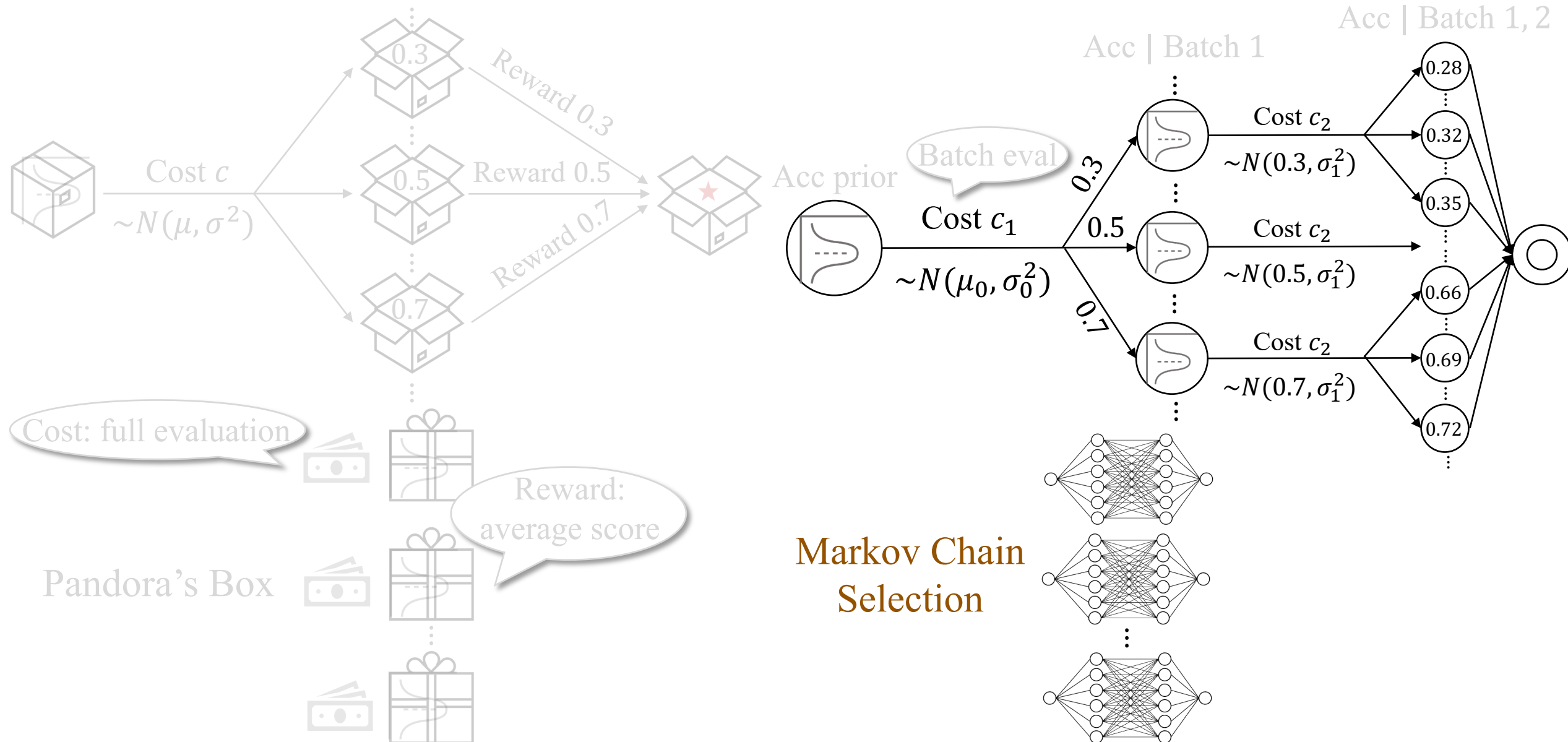
How to update posterior belief of latent mean?



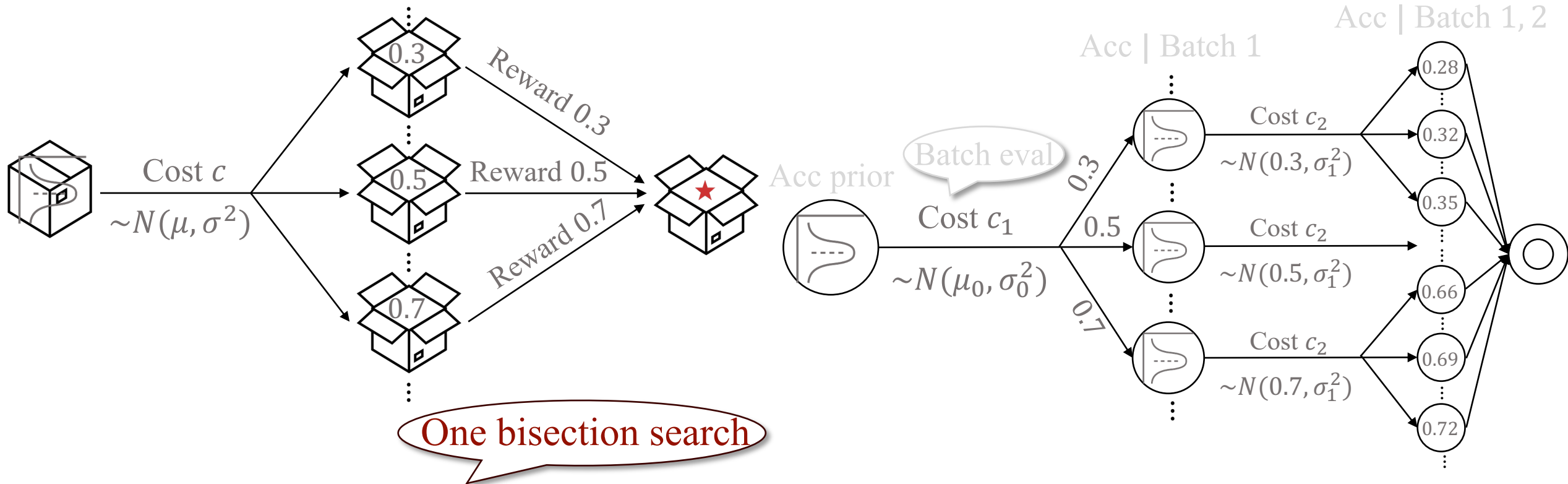
How to update posterior belief of latent mean?



How to update posterior belief of latent mean?



How to compute Gittins indices?

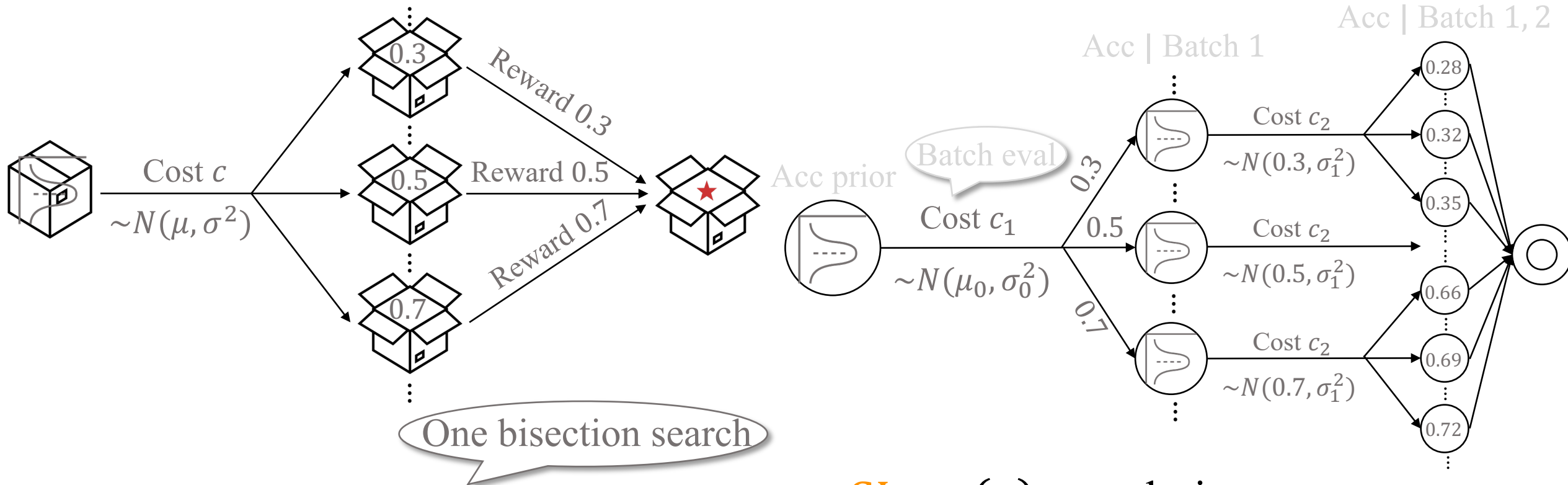


$$GI_{f|D,c}(x) := \text{solution } g \text{ s.t.}$$

$$\mathbb{E}[\max(f(x) - g, 0) | D] = c(x)$$

analytical expression
 & monotonicity in g

How to compute Gittins indices?



$$GI_{f|D,c}(x) := \text{solution } g \text{ s.t.}$$

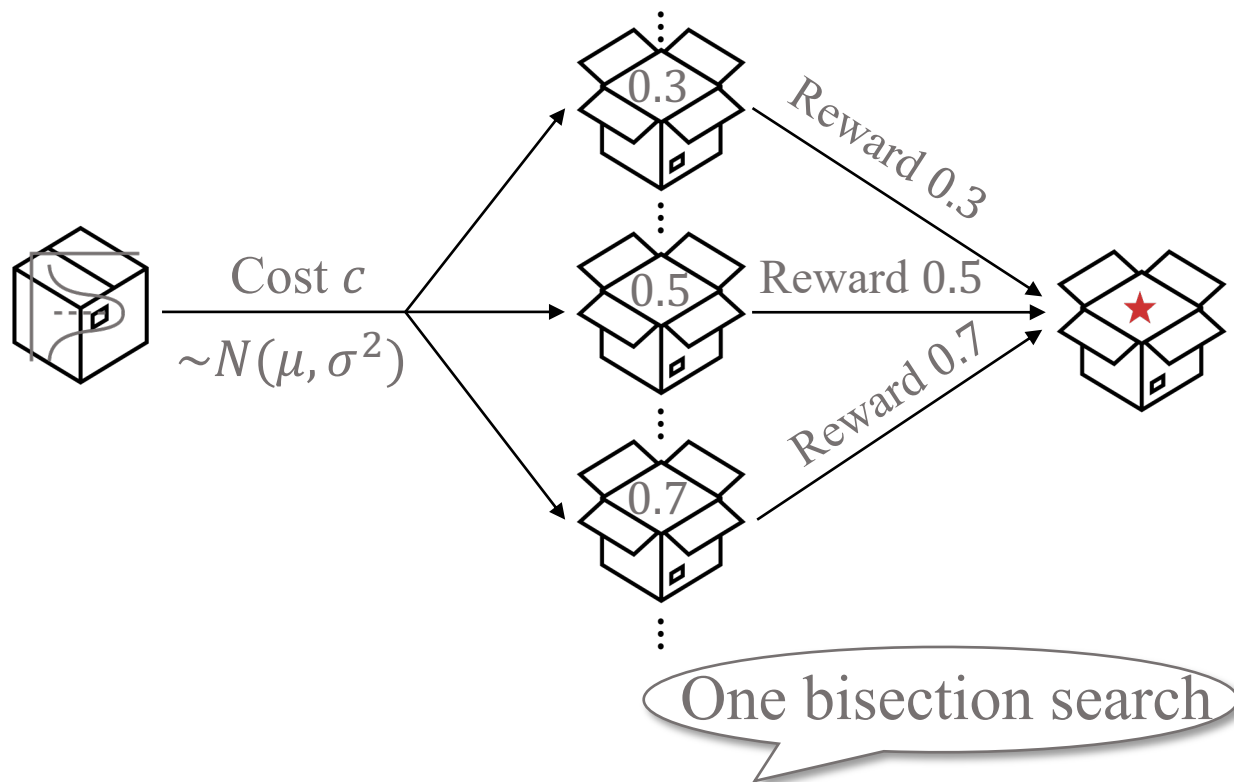
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$$GI_{f|D,c}(x) := \text{solution } g \text{ s.t.}$$

$$\mathbb{E}[\max(\mathbb{E}[\max(f(x) - g, 0) | D_1] - c_2(x), 0) | D] = c_1(x)$$

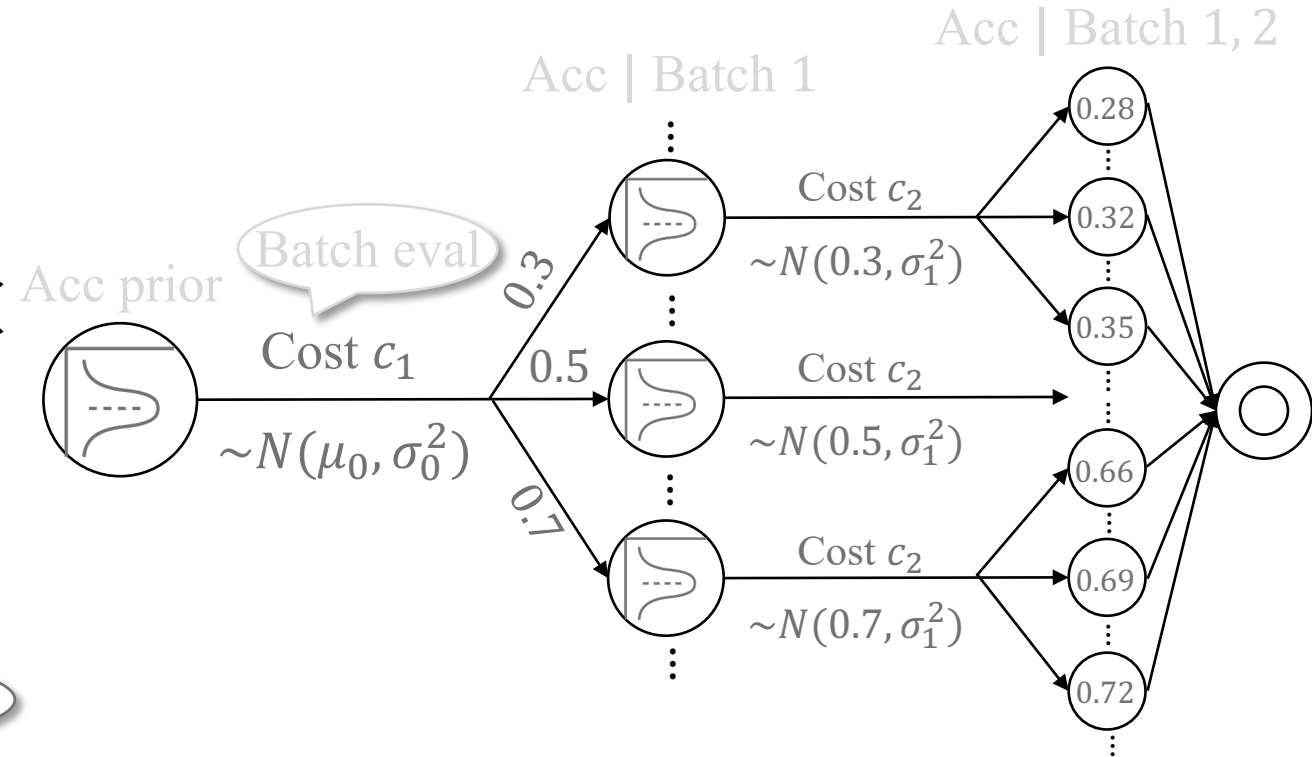
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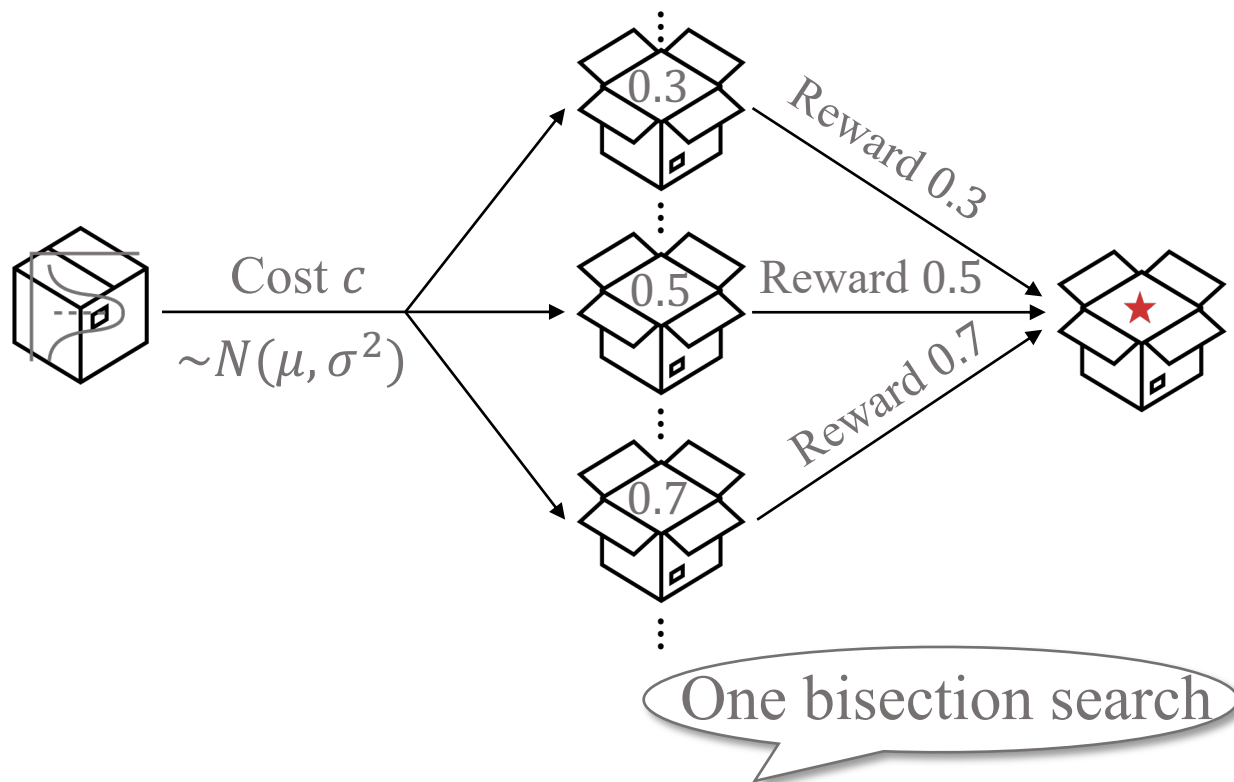


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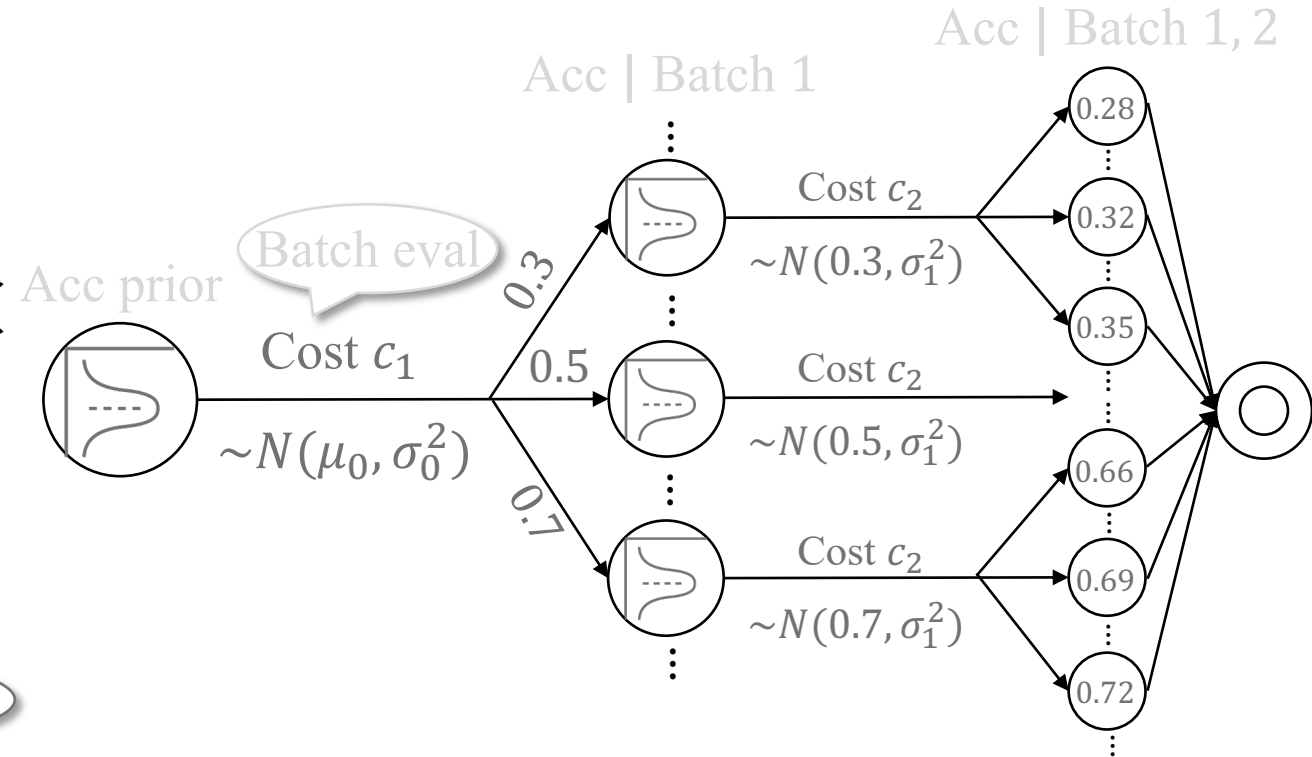
No analytical expression

How to compute Gittins indices?



$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.} \\ \mathbb{E}[\max(f(x) - g, 0) | D] = c(x)$$

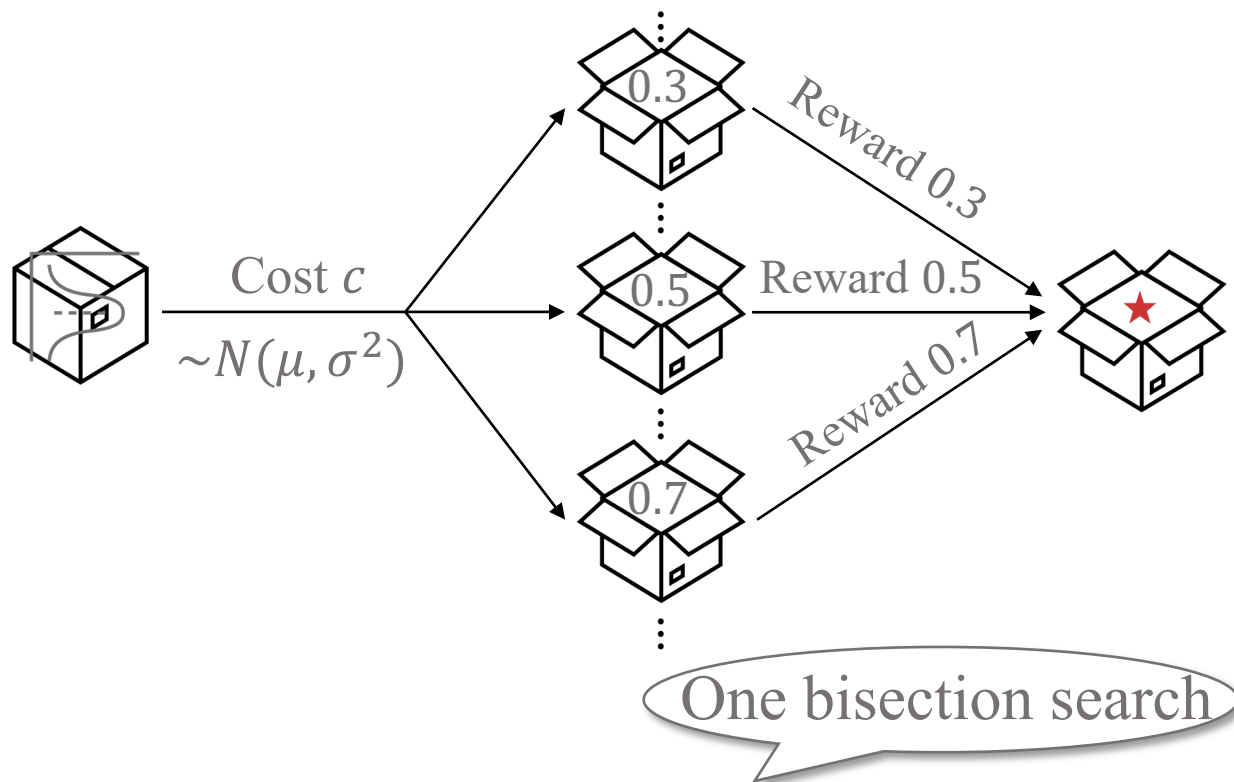
analytical expression
& monotonicity in g



$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.} \\ \mathbb{E}[\max(\mathbb{E}[\max(f(x) - g, 0) | D_1] - c_2(x), 0) | D] = c_1(x)$$

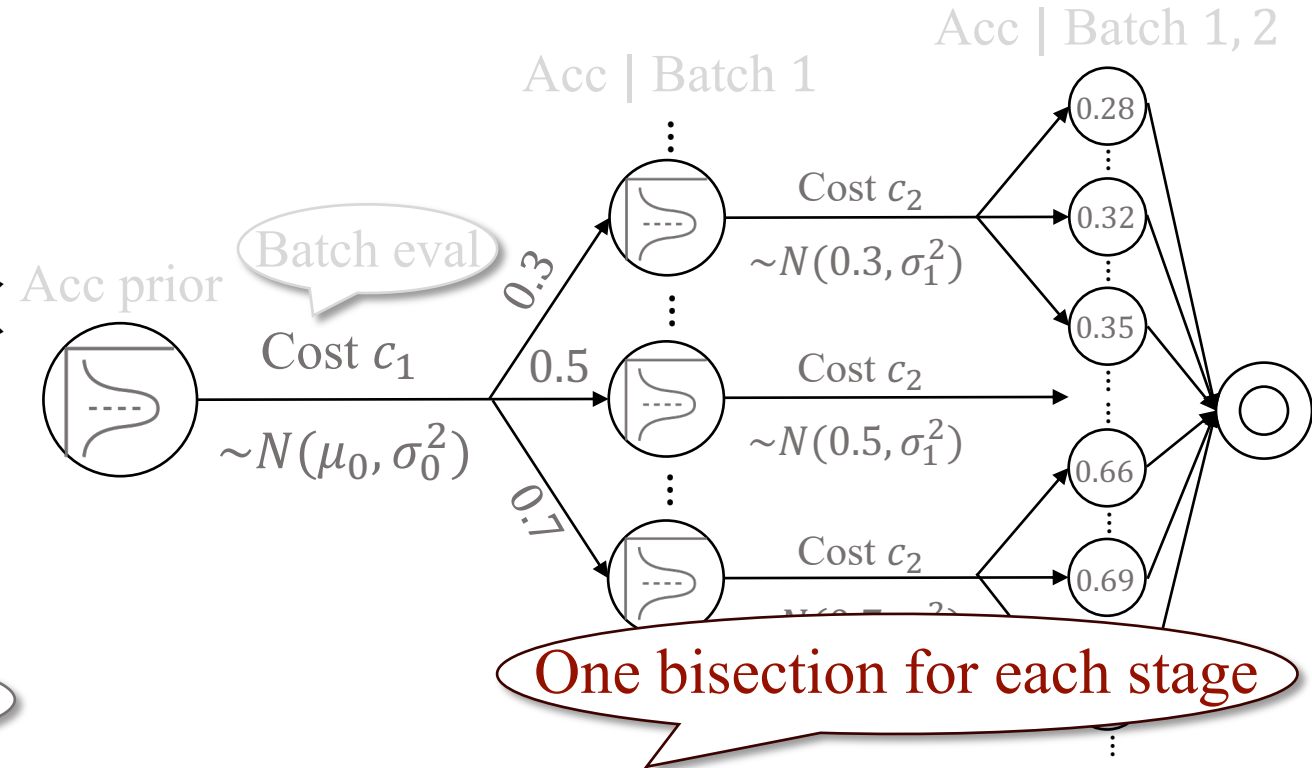
Lookup table over grids for each stage

How to compute Gittins indices?



$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.}$$

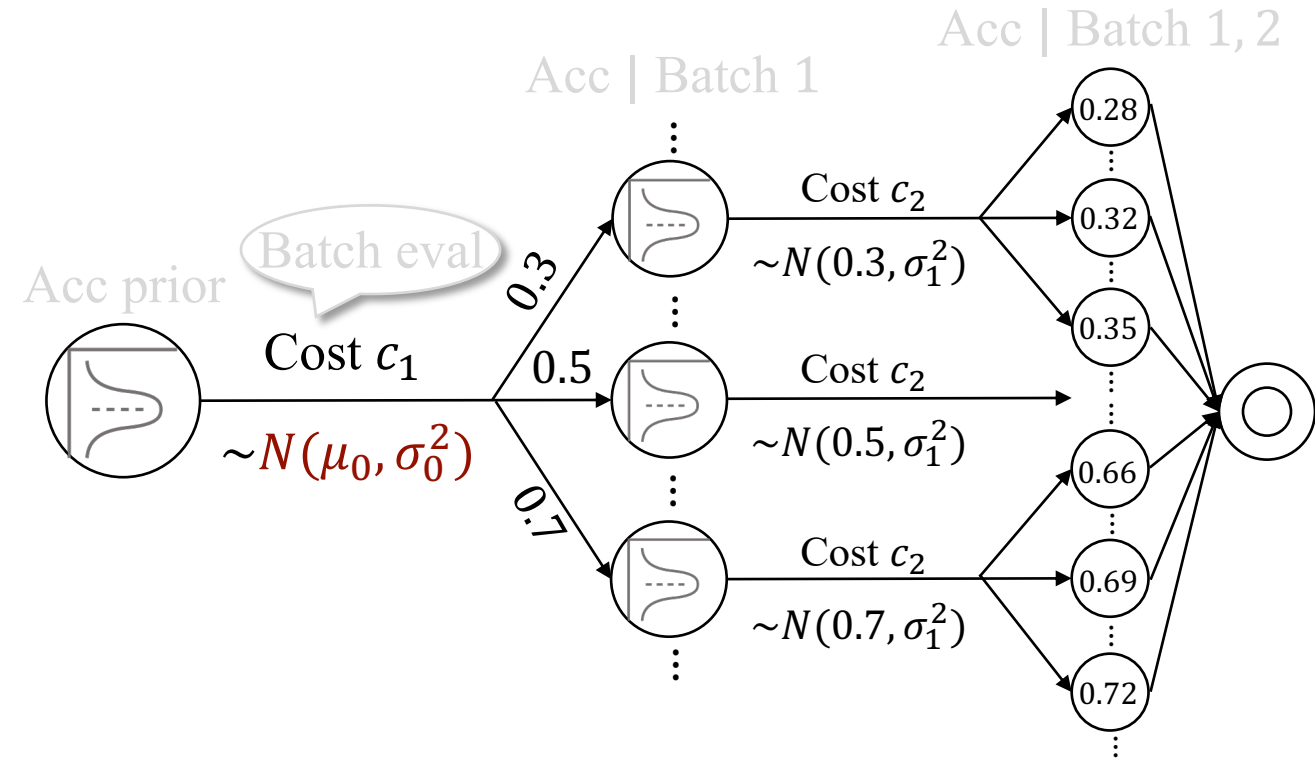
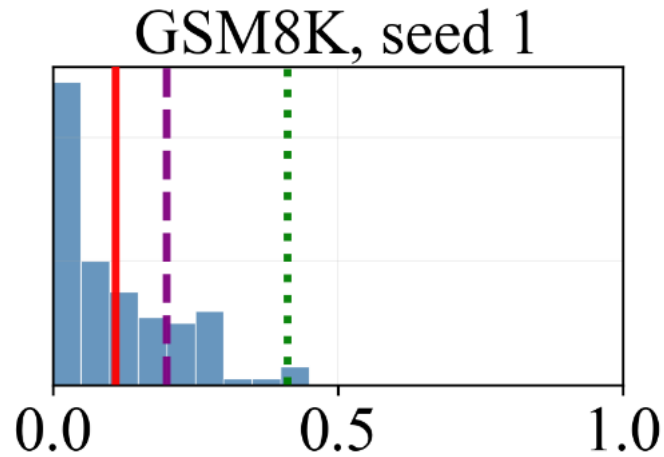
$$\underbrace{\mathbb{E}[\max(f(x) - g, 0) \mid D]}_{\text{analytical expression \& monotonicity in } g} = c(x)$$



$$\text{GI}_{f|D,c}(x) := \text{solution } g \text{ s.t.}$$

$$\underbrace{\mathbb{E}[\max(\mathbb{E}[\max(f(x) - g, 0) \mid D_1] - c_2(x), 0) \mid D]}_{\text{Lookup table over grids for each stage}} = c_1(x)$$

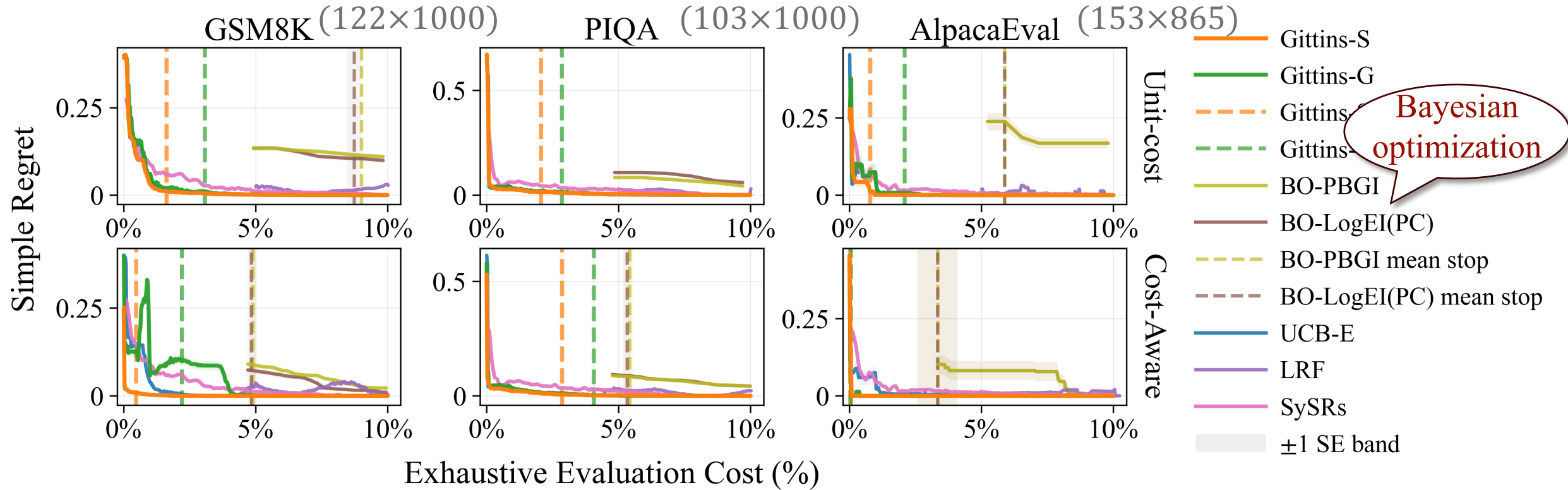
Choice of Prior



Gittins Variant	Prior	Description
Gittins-G	$N(0.5, 0.04)$	General Default
Gittins-S	e.g., $N(0.2, 0.01)$	Data-Specific

Gittins Index vs Baselines on Many-Example Tasks

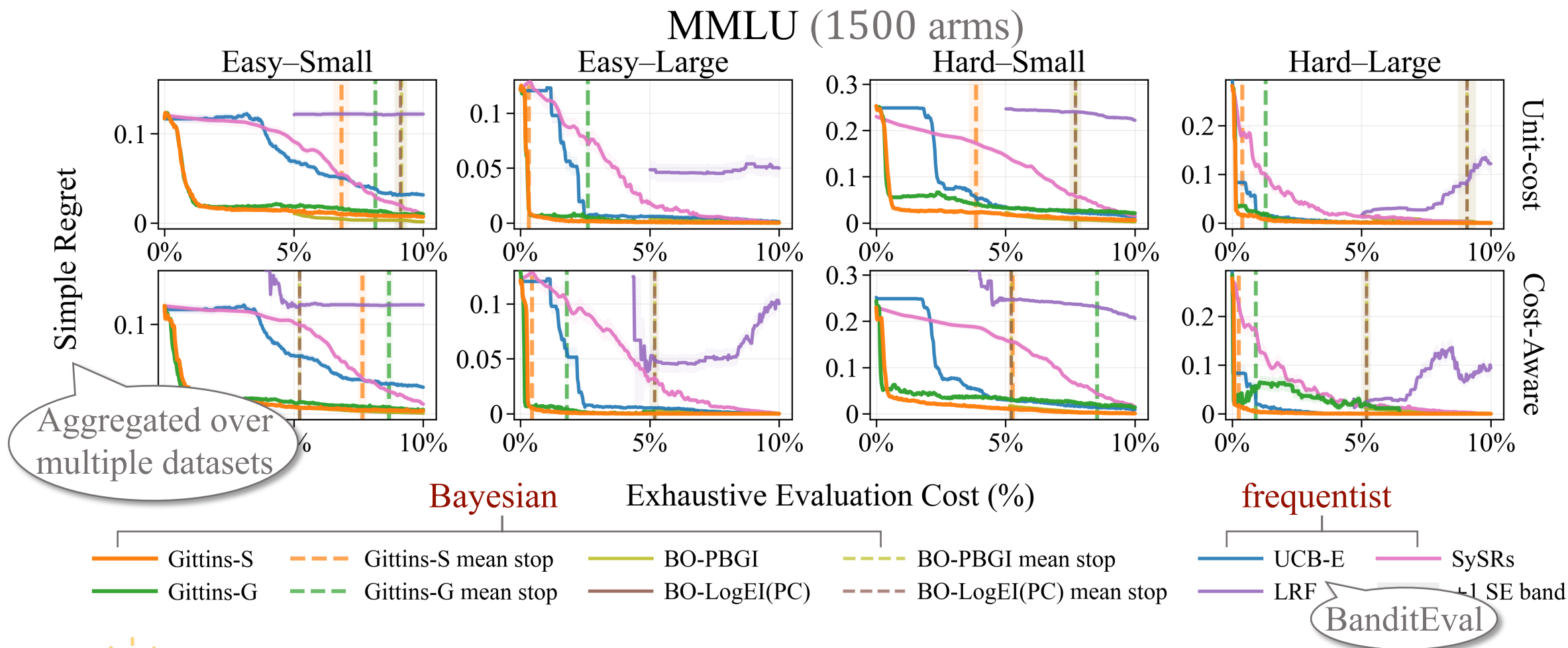
LLM evaluation:



💡 Gittins outperforms Bayesian optimization on many-example benchmarks

🔗 "Efficient Cost-Aware LLM Evaluation via Bayesian Bandit Gittins Indices." ICML'26 Workshop DEMO.

Gittins Index vs Baselines on Many-Candidate Tasks

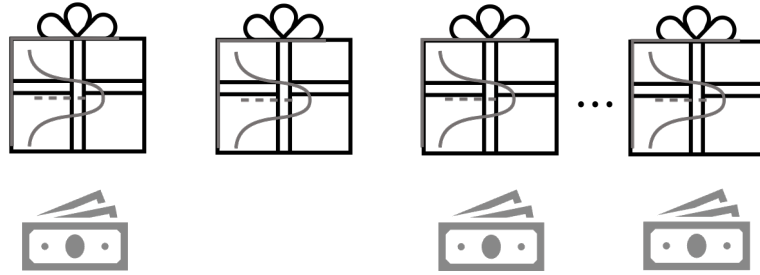


💡 Gittins outperforms frequentist methods on many-candidate tasks

○ "Efficient Cost-Aware LLM Evaluation via Bayesian
○ Bandit Gittins Indices." ICML'26 Workshop DEMO.

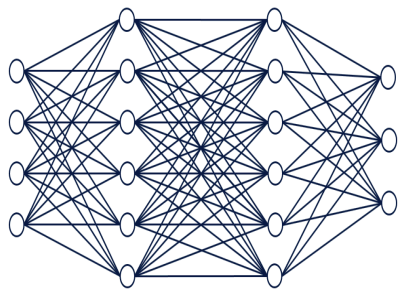
Key Contribution: Gittins Index Design Principle

Novel connection



Link to **Pandora's Box** & Markov chain selection & **Gittins index** theory

Competitive empirical performance



Model	Q1	Q2	Q3	...	avg performance
ChatGPT GPT-4o	✓	?	?	...	?
Claude 3.5 Sonnet	?	✓	?	...	?
Deepseek	✓	? Y_{ij}	?	...	?
Gemini 1.5 Pro	✗	?	✓	...	?
Llama 3.1-70B	?	✗	?	...	?
Mistral Large	?	?	✗	...	?
...	...	...	...	...	...

Applications to AutoML & **LLM**

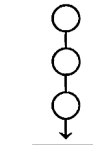
Principled decision rules



Varying evaluation costs



Adaptive stopping time

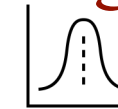


Multi-stage feedback

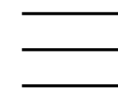
Unified framework for acquisition and stopping

Ongoing practical extensions

Part I:



Noisy observations



Batch evaluations

Part II:



Arm correlations

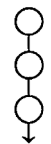


Trajectory-type feedback

My Papers Covered by This Talk



"Cost-aware Bayesian Optimization via the Pandora's Box Gittins Index." NeurIPS'24.



"Efficient Cost-Aware LLM Evaluation via Bayesian Bandit Gittins Indices." ICML'26 Workshop DEMO.



"Cost-aware Stopping for Bayesian Optimization." ICML'26.